

ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \quad (1.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (1.1b)$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \varepsilon \frac{\partial E_x}{\partial t} + \sigma E_x \quad (1.2a)$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \varepsilon \frac{\partial E_y}{\partial t} + \sigma E_y \quad (1.2b)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \varepsilon \frac{\partial E_z}{\partial t} + \sigma E_z \quad (1.2c)$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right) \quad (1.3a)$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) \quad (1.3b)$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \right) \quad (1.3c)$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i . The implementation uses

$\mathbf{E}_\alpha$: cell-centered in the α direction and nodal in the other two directions,

$\mathbf{H}_\alpha$: nodal in the α direction and cell-centered in the other two directions.

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i + \frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j + \frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k + \frac{1}{2}$
$H_{x,i,j,k}$	i	$j + \frac{1}{2}$	$k + \frac{1}{2}$
$H_{y,i,j,k}$	$i + \frac{1}{2}$	j	$k + \frac{1}{2}$
$H_{z,i,j,k}$	$i + \frac{1}{2}$	$j + \frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i + \frac{1}{2}$. They are encoded in the **MultiFab** staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j + \frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j + 1$. Therefore

$$\left. \frac{\partial E_z}{\partial y} \right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}. \quad (1.4)$$

Similarly, $E_{x,i,j,k}$ is located at j , while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j - \frac{1}{2}$ and $j + \frac{1}{2}$, so

$$\left. \frac{\partial H_z}{\partial y} \right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}. \quad (1.5)$$

This is why the magnetic update below uses forward differences such as $j + 1$ and $k + 1$, while the electric update uses backward differences such as $j - 1$ and $k - 1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update $\mathbf{H}$ by a half step, then update $\mathbf{E}$ by a full step, and finally update $\mathbf{H}$ by another half step. Define the update coefficients

$$C_a = \frac{2\varepsilon - \sigma\Delta t}{2\varepsilon + \sigma\Delta t}, \quad C_b = \frac{2\Delta t}{2\varepsilon + \sigma\Delta t}, \quad D_b = \frac{\Delta t}{\mu}, \quad (2.1)$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2} \left[\frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k+1}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta z} \right] \quad (2.2a)$$

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{2} \left[\frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} - \frac{E_{z,i+1,j,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta x} \right] \quad (2.2b)$$

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{2} \left[\frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j+1,k}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta y} \right]. \quad (2.2c)$$

In the AMReX Yee grid, with vacuum media, the magnetic field updates are implemented as

$$B_x(i, j, k) = B_x(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.3a)$$

$$B_y(i, j, k) = B_y(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.3b)$$

$$B_z(i, j, k) = B_z(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.3c)$$

2.2 Electric Full-Step

$$E_{x,i+\frac{1}{2},j,k}^{n+1} = C_{a,i+\frac{1}{2},j,k}^{n+1} E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k}^{n+1} \left[\frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.4a)$$

$$E_{y,i,j+\frac{1}{2},k}^{n+1} = C_{a,i,j+\frac{1}{2},k}^{n+1} E_{y,i,j+\frac{1}{2},k}^n + C_{b,i,j+\frac{1}{2},k}^{n+1} \left[\frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.4b)$$

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_{a,i,j,k+\frac{1}{2}}^{n+1} E_{z,i,j,k+\frac{1}{2}}^n + C_{b,i,j,k+\frac{1}{2}}^{n+1} \left[\frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.4c)$$

In the AMReX Yee grid, with vacuum media, the electric field updates are implemented as

$$E_x(i, j, k) = E_x(i, j, k) + c^2 \Delta t \left[\frac{B_z(i, j, k) - B_z(i, j - 1, k)}{\Delta y} - \frac{B_y(i, j, k) - B_y(i, j, k - 1)}{\Delta z} \right] \quad (2.5a)$$

$$E_y(i, j, k) = E_y(i, j, k) + c^2 \Delta t \left[\frac{B_x(i, j, k) - B_x(i, j, k - 1)}{\Delta z} - \frac{B_z(i, j, k) - B_z(i - 1, j, k)}{\Delta x} \right] \quad (2.5b)$$

$$E_z(i, j, k) = E_z(i, j, k) + c^2 \Delta t \left[\frac{B_y(i, j, k) - B_y(i - 1, j, k)}{\Delta x} - \frac{B_x(i, j, k) - B_x(i, j - 1, k)}{\Delta y} \right]. \quad (2.5c)$$

2.3 Second Magnetic Half-Step

$$H_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_{b, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}}{2} \left[\frac{E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} \right] \quad (2.6a)$$

$$H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_{b, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}}{2} \left[\frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} \right] \quad (2.6b)$$

$$H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} = H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}} - \frac{D_{b, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}}{2} \left[\frac{E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{y, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+1} - E_{x, i+\frac{1}{2}, j+\frac{1}{2}, k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y} \right]. \quad (2.6c)$$

In the AMReX Yee grid, with vacuum media, the second magnetic field updates are implemented as

$$B_x(i, j, k) = B_x(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_z(i, j+1, k) - E_z(i, j, k)}{\Delta y} - \frac{E_y(i, j, k+1) - E_y(i, j, k)}{\Delta z} \right] \quad (2.7a)$$

$$B_y(i, j, k) = B_y(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_x(i, j, k+1) - E_x(i, j, k)}{\Delta z} - \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x} \right] \quad (2.7b)$$

$$B_z(i, j, k) = B_z(i, j, k) - \frac{1}{2\Delta t} \left[\frac{E_y(i+1, j, k) - E_y(i, j, k)}{\Delta x} - \frac{E_x(i, j+1, k) - E_x(i, j, k)}{\Delta y} \right]. \quad (2.7c)$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \leq \frac{1}{c \sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}}}, \quad c = \frac{1}{\sqrt{\mu \varepsilon}}. \quad (2.8)$$

3 ADI Method

The ADI-FDTD update advances the fields by a full time step Δt in two half steps of size $\Delta t/2$. Unlike the explicit three-stage leapfrog update above, each ADI half step first updates $\mathbf{E}$ implicitly over $\Delta t/2$, then updates $\mathbf{H}$ explicitly over $\Delta t/2$ from the updated electric field. The first half step treats one set of implicit directions; the second half step treats the complementary directions to complete the cycle. Define the half-step update coefficients

$$C_a = \frac{4\varepsilon - \sigma\Delta t}{4\varepsilon + \sigma\Delta t}, \quad C_b = \frac{2\Delta t}{4\varepsilon + \sigma\Delta t}, \quad D_b = \frac{\Delta t}{2\mu}, \quad (3.1)$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity. These differ from the full-step coefficients in (2.1) because each electric update uses $\Delta t/2$.

3.1 First ADI Half-Step

E_x and H_z By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} - E_{x, i+\frac{1}{2}, j, k}^n}{\Delta t/2} + \sigma_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} + E_{x, i+\frac{1}{2}, j, k}^n}{2} = [H_1], \quad (3.2)$$

where

$$[H_1] = \frac{H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} - H_{z, i+\frac{1}{2}, j-\frac{1}{2}, k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^n - H_{y, i+\frac{1}{2}, j, k-\frac{1}{2}}^n}{\Delta z}. \quad (3.3)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^n + [H_1]. \quad (3.4)$$

Hence

$$E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k} [H_1], \quad (3.5)$$

By equation (1.3c),

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} \right), \quad (3.6)$$

Substituting the magnetic update into (3.5) and eliminating $H_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left[H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n}{\Delta x} \right) \right. \\ &\quad \left. - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n - D_{b,i+\frac{1}{2},j-\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n}{\Delta x} \right) \right] \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right). \end{aligned} \quad (3.7)$$

Collecting the unknown $E_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} \\ &+ \left[1 + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} \right] E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} \\ &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^n + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y \Delta x} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y \Delta x} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right). \end{aligned} \quad (3.8)$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} &= p_{x1} E_{x,i+\frac{1}{2},j,k}^n \\
&+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
&- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\
&+ \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\
&- \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right),
\end{aligned} \tag{3.9}$$

and the magnetic update can be written as

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + \frac{r_{x1}}{\Delta y} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) - \frac{r_{x1}}{\Delta x} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right). \tag{3.10}$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.11}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.12}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.13}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.14}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.15}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.16}$$

E_y and H_x By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^n}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + E_{y,i,j+\frac{1}{2},k}^n}{2} = [H_2], \tag{3.17}$$

where

$$[H_2] = \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n}{\Delta x}. \tag{3.18}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} \right) E_y^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} \right) E_y^n + [H_2]. \tag{3.19}$$

Hence

$$E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} = C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^n + C_{b,i,j+\frac{1}{2},k} [H_2], \tag{3.20}$$

By equation (1.3a),

$$\begin{aligned}
H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} &= H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} \right. \\
&\quad \left. - \frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} \right),
\end{aligned} \tag{3.21}$$

Substituting the magnetic update into (3.20) and eliminating $H_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned}
E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} &= C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^n \\
&+ \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left[H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta y} \right) \right. \\
&\quad \left. - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^n - D_{b,i,j+\frac{1}{2},k-\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k-\frac{1}{2}}^n - E_{z,i,j,k-\frac{1}{2}}^n}{\Delta y} \right) \right] \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n \right).
\end{aligned} \tag{3.22}$$

Collecting the unknown $E_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned}
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} \\
&+ \left[1 + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} \right] E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}} \\
&= C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^n + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^n \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z \Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right) \\
&+ \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z \Delta y} \left(E_{z,i,j+1,k-\frac{1}{2}}^n - E_{z,i,j,k-\frac{1}{2}}^n \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n \right).
\end{aligned} \tag{3.23}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{y1} E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}} + \beta_{y1} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - \gamma_{y1} E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} &= p_{y1} E_{y,i,j+\frac{1}{2},k}^n \\
&+ \frac{1}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^n \right) \\
&- \frac{1}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n \right) \\
&+ \frac{q_{y1}}{\Delta y \Delta z} \left(E_{z,i,j+1,k-\frac{1}{2}}^n - E_{z,i,j,k-\frac{1}{2}}^n \right) \\
&- \frac{r_{y1}}{\Delta y \Delta z} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right),
\end{aligned} \tag{3.24}$$

and the magnetic update can be written as

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n + \frac{r_{y1}}{\Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) - \frac{r_{y1}}{\Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right). \tag{3.25}$$

where

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}, \quad (3.26)$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \quad (3.27)$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2}, \quad (3.28)$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \quad (3.29)$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \quad (3.30)$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \quad (3.31)$$

E_z and H_y By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^n}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + E_{z,i,j,k+\frac{1}{2}}^n}{2} = [H_3], \quad (3.32)$$

where

$$[H_3] = \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n}{\Delta y}. \quad (3.33)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} \right) E_z^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} \right) E_z^n + [H_3]. \quad (3.34)$$

Hence

$$E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^n + C_{b,i,j,k+\frac{1}{2}} [H_3], \quad (3.35)$$

By equation (1.3b),

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} \right), \quad (3.36)$$

Substituting the magnetic update into (3.35) and eliminating $H_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} &= C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^n \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left[H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n}{\Delta z} \right) \right. \\ &\quad \left. - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^n - D_{b,i-\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i-\frac{1}{2},j,k+1}^n - E_{x,i-\frac{1}{2},j,k}^n}{\Delta z} \right) \right] \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n \right). \end{aligned} \quad (3.37)$$

Collecting the unknown $E_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned}
& - \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
& + \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} \right] E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
& - \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
& = C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^n + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^n \right) \\
& - \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x \Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right) \\
& + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x \Delta z} \left(E_{x,i-\frac{1}{2},j,k+1}^n - E_{x,i-\frac{1}{2},j,k}^n \right) \\
& - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n \right).
\end{aligned} \tag{3.38}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
& -\alpha_{z1} E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \beta_{z1} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \gamma_{z1} E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = p_{z1} E_{z,i,j,k+\frac{1}{2}}^n \\
& + \frac{1}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^n \right) \\
& - \frac{1}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n \right) \\
& + \frac{q_{z1}}{\Delta z \Delta x} \left(E_{x,i-\frac{1}{2},j,k+1}^n - E_{x,i-\frac{1}{2},j,k}^n \right) \\
& - \frac{r_{z1}}{\Delta z \Delta x} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right),
\end{aligned} \tag{3.39}$$

and the magnetic update can be written as

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n + \frac{r_{z1}}{\Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) - \frac{r_{z1}}{\Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right). \tag{3.40}$$

where

$$\alpha_{z1} = \frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.41}$$

$$\beta_{z1} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z1} + \gamma_{z1}, \tag{3.42}$$

$$\gamma_{z1} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.43}$$

$$p_{z1} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.44}$$

$$q_{z1} = D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{3.45}$$

$$r_{z1} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.46}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$\begin{aligned}
-E_x^{n+\frac{1}{2}}(i, j-1, k) + \left(2 + \frac{4\Delta y^2}{c^2\Delta t^2}\right) E_x^{n+\frac{1}{2}}(i, j, k) - E_x^{n+\frac{1}{2}}(i, j+1, k) &= \frac{4\Delta y^2}{c^2\Delta t^2} E_x^n(i, j, k) \\
&+ \frac{2\Delta y^2}{\Delta t} \left[\frac{B_z^n(i, j, k) - B_z^n(i, j-1, k)}{\Delta y} - \frac{B_y^n(i, j, k) - B_y^n(i, j, k-1)}{\Delta z} \right] \\
&+ \Delta y \left[\frac{E_y^n(i+1, j-1, k) - E_y^n(i, j-1, k)}{\Delta x} - \frac{E_y^n(i+1, j, k) - E_y^n(i, j, k)}{\Delta x} \right], \tag{3.47a}
\end{aligned}$$

$$\begin{aligned}
-E_y^{n+\frac{1}{2}}(i, j, k-1) + \left(2 + \frac{4\Delta z^2}{c^2\Delta t^2}\right) E_y^{n+\frac{1}{2}}(i, j, k) - E_y^{n+\frac{1}{2}}(i, j, k+1) &= \frac{4\Delta z^2}{c^2\Delta t^2} E_y^n(i, j, k) \\
&+ \frac{2\Delta z^2}{\Delta t} \left[\frac{B_x^n(i, j, k) - B_x^n(i, j, k-1)}{\Delta z} - \frac{B_z^n(i, j, k) - B_z^n(i-1, j, k)}{\Delta x} \right] \\
&+ \Delta z \left[\frac{E_z^n(i, j+1, k-1) - E_z^n(i, j, k-1)}{\Delta y} - \frac{E_z^n(i, j+1, k) - E_z^n(i, j, k)}{\Delta y} \right], \tag{3.47b}
\end{aligned}$$

$$\begin{aligned}
-E_z^{n+\frac{1}{2}}(i-1, j, k) + \left(2 + \frac{4\Delta x^2}{c^2\Delta t^2}\right) E_z^{n+\frac{1}{2}}(i, j, k) - E_z^{n+\frac{1}{2}}(i+1, j, k) &= \frac{4\Delta x^2}{c^2\Delta t^2} E_z^n(i, j, k) \\
&+ \frac{2\Delta x^2}{\Delta t} \left[\frac{B_y^n(i, j, k) - B_y^n(i-1, j, k)}{\Delta x} - \frac{B_x^n(i, j, k) - B_x^n(i, j-1, k)}{\Delta y} \right] \\
&+ \Delta x \left[\frac{E_x^n(i-1, j, k+1) - E_x^n(i-1, j, k)}{\Delta z} - \frac{E_x^n(i, j, k+1) - E_x^n(i, j, k)}{\Delta z} \right], \tag{3.47c}
\end{aligned}$$

$$B_x^{n+\frac{1}{2}}(i, j, k) = B_x^n(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_y^{n+\frac{1}{2}}(i, j, k+1) - E_y^{n+\frac{1}{2}}(i, j, k)}{\Delta z} - \frac{E_z^n(i, j+1, k) - E_z^n(i, j, k)}{\Delta y} \right], \tag{3.47d}$$

$$B_y^{n+\frac{1}{2}}(i, j, k) = B_y^n(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_z^{n+\frac{1}{2}}(i+1, j, k) - E_z^{n+\frac{1}{2}}(i, j, k)}{\Delta x} - \frac{E_x^n(i, j, k+1) - E_x^n(i, j, k)}{\Delta z} \right], \tag{3.47e}$$

$$B_z^{n+\frac{1}{2}}(i, j, k) = B_z^n(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_x^{n+\frac{1}{2}}(i, j+1, k) - E_x^{n+\frac{1}{2}}(i, j, k)}{\Delta y} - \frac{E_y^n(i+1, j, k) - E_y^n(i, j, k)}{\Delta x} \right]. \tag{3.47f}$$

3.2 Second ADI Half-Step

E_x and H_y By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+1} - E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}}{\Delta t/2} + \sigma_{i+\frac{1}{2}, j, k} \frac{E_{x, i+\frac{1}{2}, j, k}^{n+1} + E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}}{2} = [H_4], \tag{3.48}$$

where

$$[H_4] = \frac{H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^{n+\frac{1}{2}} - H_{z, i+\frac{1}{2}, j-\frac{1}{2}, k}^{n+\frac{1}{2}}}{\Delta y} - \frac{H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^{n+1} - H_{y, i+\frac{1}{2}, j, k-\frac{1}{2}}^{n+1}}{\Delta z}. \tag{3.49}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} + [H_4]. \tag{3.50}$$

Hence

$$E_{x, i+\frac{1}{2}, j, k}^{n+1} = C_{a, i+\frac{1}{2}, j, k} E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} + C_{b, i+\frac{1}{2}, j, k} [H_4], \tag{3.51}$$

By equation (1.3b),

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1}}{\Delta z} \right), \quad (3.52)$$

Substituting the magnetic update into (3.51) and eliminating H_y^{n+1} terms gives

$$\begin{aligned} E_{x,i+\frac{1}{2},j,k}^{n+1} &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left[H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1}}{\Delta z} \right) \right. \\ &\quad \left. - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} - D_{b,i+\frac{1}{2},j,k-\frac{1}{2}} \left(\frac{E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{E_{x,i+\frac{1}{2},j,k}^{n+1} - E_{x,i+\frac{1}{2},j,k-1}^{n+1}}{\Delta z} \right) \right]. \end{aligned} \quad (3.53)$$

Collecting the unknown E_x^{n+1} terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} E_{x,i+\frac{1}{2},j,k+1}^{n+1} \\ &+ \left[1 + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2} \right] E_{x,i+\frac{1}{2},j,k}^{n+1} \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2} E_{x,i+\frac{1}{2},j,k-1}^{n+1} \\ &= C_{a,i+\frac{1}{2},j,k} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z \Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z \Delta x} \left(E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right). \end{aligned} \quad (3.54)$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned} -\alpha_{x2} E_{x,i+\frac{1}{2},j,k-1}^{n+1} + \beta_{x2} E_{x,i+\frac{1}{2},j,k}^{n+1} - \gamma_{x2} E_{x,i+\frac{1}{2},j,k+1}^{n+1} &= p_{x2} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\ &+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &+ \frac{q_{x2}}{\Delta x \Delta z} \left(E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{r_{x2}}{\Delta x \Delta z} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right), \end{aligned} \quad (3.55)$$

and the magnetic update can be written as

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} = H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{r_{x2}}{\Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) - \frac{r_{x2}}{\Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1} \right). \quad (3.56)$$

where

$$\alpha_{x2} = \frac{D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}, \quad (3.57)$$

$$\beta_{x2} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x2} + \gamma_{x2}, \quad (3.58)$$

$$\gamma_{x2} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2}, \quad (3.59)$$

$$p_{x2} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \quad (3.60)$$

$$q_{x2} = D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}, \quad (3.61)$$

$$r_{x2} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \quad (3.62)$$

E_y and H_z By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E_{y,i,j+\frac{1}{2},k}^{n+1} + E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{2} = [H_5], \quad (3.63)$$

where

$$[H_5] = \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z} - \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+1}}{\Delta x}. \quad (3.64)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} \right) E_y^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} \right) E_y^{n+\frac{1}{2}} + [H_5]. \quad (3.65)$$

Hence

$$E_{y,i,j+\frac{1}{2},k}^{n+1} = C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + C_{b,i,j+\frac{1}{2},k} [H_5], \quad (3.66)$$

By equation (1.3c),

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta x} \right), \quad (3.67)$$

Substituting the magnetic update into (3.66) and eliminating H_z^{n+1} terms gives

$$\begin{aligned} E_{y,i,j+\frac{1}{2},k}^{n+1} &= C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \\ &+ \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left[H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta x} \right) \right. \\ &\quad \left. - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - D_{b,i-\frac{1}{2},j+\frac{1}{2},k} \left(\frac{E_{x,i-\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i-\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y} - \frac{E_{y,i,j+\frac{1}{2},k}^{n+1} - E_{y,i-1,j+\frac{1}{2},k}^{n+1}}{\Delta x} \right) \right]. \end{aligned} \quad (3.68)$$

Collecting the unknown E_y^{n+1} terms gives

$$\begin{aligned}
& -\frac{C_{b,i,j+\frac{1}{2},k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} E_{y,i+1,j+\frac{1}{2},k}^{n+1} \\
& + \left[1 + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} \right] E_{y,i,j+\frac{1}{2},k}^{n+1} \\
& - \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2} E_{y,i-1,j+\frac{1}{2},k}^{n+1} \\
& = C_{a,i,j+\frac{1}{2},k} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
& - \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
& - \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x \Delta y} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\
& + \frac{C_{b,i,j+\frac{1}{2},k} D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x \Delta y} \left(E_{x,i-\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i-\frac{1}{2},j,k}^{n+\frac{1}{2}} \right).
\end{aligned} \tag{3.69}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{y2} E_{y,i-1,j+\frac{1}{2},k}^{n+1} + \beta_{y2} E_{y,i,j+\frac{1}{2},k}^{n+1} - \gamma_{y2} E_{y,i+1,j+\frac{1}{2},k}^{n+1} & = p_{y2} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \\
& + \frac{1}{\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
& - \frac{1}{\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
& + \frac{q_{y2}}{\Delta y \Delta x} \left(E_{x,i-\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i-\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\
& - \frac{r_{y2}}{\Delta y \Delta x} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right),
\end{aligned} \tag{3.70}$$

and the magnetic update can be written as

$$H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} = H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + \frac{r_{y2}}{\Delta y} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) - \frac{r_{y2}}{\Delta x} \left(E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1} \right). \tag{3.71}$$

where

$$\alpha_{y2} = \frac{D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.72}$$

$$\beta_{y2} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y2} + \gamma_{y2}, \tag{3.73}$$

$$\gamma_{y2} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.74}$$

$$p_{y2} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.75}$$

$$q_{y2} = D_{b,i-\frac{1}{2},j+\frac{1}{2},k}, \tag{3.76}$$

$$r_{y2} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.77}$$

E_z and H_x By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E_{z,i,j,k+\frac{1}{2}}^{n+1} + E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{2} = [H_6], \tag{3.78}$$

where

$$[H_6] = \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x} - \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+1}}{\Delta y}. \quad (3.79)$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} + [H_6]. \quad (3.80)$$

Hence

$$E_{z,i,j,k+\frac{1}{2}}^{n+1} = C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}} [H_6], \quad (3.81)$$

By equation (1.3a),

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta y} \right), \quad (3.82)$$

Substituting the magnetic update into (3.81) and eliminating H_x^{n+1} terms gives

$$\begin{aligned} E_{z,i,j,k+\frac{1}{2}}^{n+1} &= C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left[H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta y} \right) \right. \\ &\quad \left. - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - D_{b,i,j-\frac{1}{2},k+\frac{1}{2}} \left(\frac{E_{y,i,j-\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z} - \frac{E_{z,i,j,k+\frac{1}{2}}^{n+1} - E_{z,i,j-1,k+\frac{1}{2}}^{n+1}}{\Delta y} \right) \right]. \end{aligned} \quad (3.83)$$

Collecting the unknown E_z^{n+1} terms gives

$$\begin{aligned} &- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} E_{z,i,j+1,k+\frac{1}{2}}^{n+1} \\ &+ \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} + \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} \right] E_{z,i,j,k+\frac{1}{2}}^{n+1} \\ &- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} E_{z,i,j-1,k+\frac{1}{2}}^{n+1} \\ &= C_{a,i,j,k+\frac{1}{2}} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y \Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}} D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y \Delta z} \left(E_{y,i,j-\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j-\frac{1}{2},k}^{n+\frac{1}{2}} \right). \end{aligned} \quad (3.84)$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{z2} E_{z,i,j-1,k+\frac{1}{2}}^{n+1} + \beta_{z2} E_{z,i,j,k+\frac{1}{2}}^{n+1} - \gamma_{z2} E_{z,i,j+1,k+\frac{1}{2}}^{n+1} &= p_{z2} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\
&+ \frac{1}{\Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&- \frac{1}{\Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&+ \frac{q_{z2}}{\Delta z \Delta y} \left(E_{y,i,j-\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
&- \frac{r_{z2}}{\Delta z \Delta y} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right),
\end{aligned} \tag{3.85}$$

and the magnetic update can be written as

$$H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{r_{z2}}{\Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) - \frac{r_{z2}}{\Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1} \right). \tag{3.86}$$

where

$$\alpha_{z2} = \frac{D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.87}$$

$$\beta_{z2} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z2} + \gamma_{z2}, \tag{3.88}$$

$$\gamma_{z2} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.89}$$

$$p_{z2} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.90}$$

$$q_{z2} = D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}, \tag{3.91}$$

$$r_{z2} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.92}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$\begin{aligned}
-E_x^{n+1}(i, j, k-1) + \left(2 + \frac{4\Delta z^2}{c^2 \Delta t^2} \right) E_x^{n+1}(i, j, k) - E_x^{n+1}(i, j, k+1) &= \frac{4\Delta z^2}{c^2 \Delta t^2} E_x^{n+\frac{1}{2}}(i, j, k) \\
&+ \frac{2\Delta z^2}{\Delta t} \left[\frac{B_z^{n+\frac{1}{2}}(i, j, k) - B_z^{n+\frac{1}{2}}(i, j-1, k)}{\Delta y} - \frac{B_y^{n+\frac{1}{2}}(i, j, k) - B_y^{n+\frac{1}{2}}(i, j, k-1)}{\Delta z} \right] \\
&+ \Delta z \left[\frac{E_z^{n+\frac{1}{2}}(i+1, j, k-1) - E_z^{n+\frac{1}{2}}(i, j, k-1)}{\Delta x} - \frac{E_z^{n+\frac{1}{2}}(i+1, j, k) - E_z^{n+\frac{1}{2}}(i, j, k)}{\Delta x} \right],
\end{aligned} \tag{3.93a}$$

$$\begin{aligned}
-E_y^{n+1}(i-1, j, k) + \left(2 + \frac{4\Delta x^2}{c^2 \Delta t^2} \right) E_y^{n+1}(i, j, k) - E_y^{n+1}(i+1, j, k) &= \frac{4\Delta x^2}{c^2 \Delta t^2} E_y^{n+\frac{1}{2}}(i, j, k) \\
&+ \frac{2\Delta x^2}{\Delta t} \left[\frac{B_x^{n+\frac{1}{2}}(i, j, k) - B_x^{n+\frac{1}{2}}(i, j, k-1)}{\Delta z} - \frac{B_z^{n+\frac{1}{2}}(i, j, k) - B_z^{n+\frac{1}{2}}(i-1, j, k)}{\Delta x} \right] \\
&+ \Delta x \left[\frac{E_x^{n+\frac{1}{2}}(i-1, j+1, k) - E_x^{n+\frac{1}{2}}(i-1, j, k)}{\Delta y} - \frac{E_x^{n+\frac{1}{2}}(i, j+1, k) - E_x^{n+\frac{1}{2}}(i, j, k)}{\Delta y} \right],
\end{aligned} \tag{3.93b}$$

$$\begin{aligned}
-E_z^{n+1}(i, j-1, k) + \left(2 + \frac{4\Delta y^2}{c^2\Delta t^2}\right) E_z^{n+1}(i, j, k) - E_z^{n+1}(i, j+1, k) &= \frac{4\Delta y^2}{c^2\Delta t^2} E_z^{n+\frac{1}{2}}(i, j, k) \\
&+ \frac{2\Delta y^2}{\Delta t} \left[\frac{B_y^{n+\frac{1}{2}}(i, j, k) - B_y^{n+\frac{1}{2}}(i-1, j, k)}{\Delta x} - \frac{B_x^{n+\frac{1}{2}}(i, j, k) - B_x^{n+\frac{1}{2}}(i, j-1, k)}{\Delta y} \right] \\
&+ \Delta y \left[\frac{E_y^{n+\frac{1}{2}}(i, j-1, k+1) - E_y^{n+\frac{1}{2}}(i, j-1, k)}{\Delta z} - \frac{E_y^{n+\frac{1}{2}}(i, j, k+1) - E_y^{n+\frac{1}{2}}(i, j, k)}{\Delta z} \right],
\end{aligned} \tag{3.93c}$$

$$B_x^{n+1}(i, j, k) = B_x^{n+\frac{1}{2}}(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_y^{n+\frac{1}{2}}(i, j, k+1) - E_y^{n+\frac{1}{2}}(i, j, k)}{\Delta z} - \frac{E_z^{n+1}(i, j+1, k) - E_z^{n+1}(i, j, k)}{\Delta y} \right], \tag{3.93d}$$

$$B_y^{n+1}(i, j, k) = B_y^{n+\frac{1}{2}}(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_z^{n+\frac{1}{2}}(i+1, j, k) - E_z^{n+\frac{1}{2}}(i, j, k)}{\Delta x} - \frac{E_x^{n+1}(i, j, k+1) - E_x^{n+1}(i, j, k)}{\Delta z} \right], \tag{3.93e}$$

$$B_z^{n+1}(i, j, k) = B_z^{n+\frac{1}{2}}(i, j, k) + \frac{\Delta t}{2} \left[\frac{E_x^{n+\frac{1}{2}}(i, j+1, k) - E_x^{n+\frac{1}{2}}(i, j, k)}{\Delta y} - \frac{E_y^{n+1}(i+1, j, k) - E_y^{n+1}(i, j, k)}{\Delta x} \right]. \tag{3.93f}$$

4 Solution of Linear Systems

See [Tridiagonal matrix algorithm](#) for more details.

5 Perfect Electric Conductor Boundary Conditions

For a perfect electric conductor (PEC) boundary with outward unit normal $\hat{\mathbf{n}}$, the electric field satisfies

$$\hat{\mathbf{n}} \times \mathbf{E} = 0. \tag{5.1}$$

That is, the tangential electric field vanishes on the conducting surface. If the field inside the conductor is not evolved and the initial data are consistent with the boundary, then Faraday's law also implies that the normal magnetic flux remains zero on the wall,

$$\hat{\mathbf{n}} \cdot \mathbf{B} = 0. \tag{5.2}$$

Now consider a rectangular domain with PEC boundaries on the two faces normal to x , namely the planes $x = x_{\text{lo}}$ and $x = x_{\text{hi}}$, while the four faces normal to y and z remain periodic. On the AMReX Yee grid, E_y and E_z are nodal in x and therefore live directly on the x -boundary planes, whereas E_x is centered at $i + \frac{1}{2}$ in the x direction and is not stored on the boundary itself. If the domain has N_x cells in x , the discrete PEC conditions are therefore

$$E_y(0, j, k) = E_y(N_x, j, k) = 0, \quad E_z(0, j, k) = E_z(N_x, j, k) = 0. \tag{5.3}$$

No value is imposed directly on E_x at the wall, since it is the normal component.

In the Yee update, the implementation is to overwrite the tangential electric fields on the two PEC planes after each electric-field stage,

$$E_y^m(0, j, k) = E_y^m(N_x, j, k) = 0, \quad E_z^m(0, j, k) = E_z^m(N_x, j, k) = 0, \tag{5.4}$$

where $m = n + \frac{1}{2}$ or $n + 1$ depending on the stage. The neighboring magnetic updates then use differences against these zero boundary values. For example, the x -derivative appearing in the B_y update is approximated by

$$\left. \frac{\partial E_z}{\partial x} \right|_{i+\frac{1}{2}, j, k} \approx \frac{E_z(i+1, j, k) - E_z(i, j, k)}{\Delta x}. \tag{5.5}$$

At the first interior face next to $x = x_{\text{lo}}$,

$$\left. \frac{\partial E_z}{\partial x} \right|_{\frac{1}{2}, j, k} \approx \frac{E_z(1, j, k) - E_z(0, j, k)}{\Delta x} = \frac{E_z(1, j, k)}{\Delta x}, \tag{5.6}$$

while at the last interior face next to $x = x_{\text{hi}}$,

$$\left. \frac{\partial E_z}{\partial x} \right|_{N_x-\frac{1}{2}, j, k} \approx \frac{E_z(N_x, j, k) - E_z(N_x-1, j, k)}{\Delta x} = -\frac{E_z(N_x-1, j, k)}{\Delta x}. \tag{5.7}$$

Likewise, the update for the normal magnetic component B_x on the PEC planes involves only tangential derivatives of E_y and E_z , so if B_x is initialized consistently on the wall then it remains zero there.

In the ADI scheme, the same PEC condition enters as a Dirichlet condition whenever a tangential electric component is solved implicitly along x . Because the y and z directions remain periodic, the implicit solves along those directions are unchanged and stay cyclic. The only modified line solves are the x -directed ones:

- the E_z solve in the first half-step, equation (3.39);
- the E_y solve in the second half-step, equation (3.70).

For the first half-step, the E_z solve uses

$$E_{z,0,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = 0, \quad E_{z,N_x,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = 0. \quad (5.8)$$

Hence the first and last interior tridiagonal equations are

$$\beta_{z1} E_{z,1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \gamma_{z1} E_{z,2,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = \text{RHS}, \quad (5.9)$$

and

$$-\alpha_{z1} E_{z,N_x-2,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \beta_{z1} E_{z,N_x-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = \text{RHS}, \quad (5.10)$$

respectively. The boundary unknowns do not participate in the solve; they are prescribed by the PEC condition.

Similarly, in the second half-step the E_y solve uses

$$E_{y,0,j+\frac{1}{2},k}^{n+1} = 0, \quad E_{y,N_x,j+\frac{1}{2},k}^{n+1} = 0. \quad (5.11)$$

Hence the first and last interior tridiagonal equations are

$$\beta_{y2} E_{y,1,j+\frac{1}{2},k}^{n+1} - \gamma_{y2} E_{y,2,j+\frac{1}{2},k}^{n+1} = \text{RHS}, \quad (5.12)$$

and

$$-\alpha_{y2} E_{y,N_x-2,j+\frac{1}{2},k}^{n+1} + \beta_{y2} E_{y,N_x-1,j+\frac{1}{2},k}^{n+1} = \text{RHS}, \quad (5.13)$$

respectively.

Note that we do not need to explicitly impose $\hat{\mathbf{n}} \cdot \mathbf{B} = 0$ separately. For an x -normal wall, the magnetic update on the boundary plane $i = i_b$, with $i_b \in \{0, N_x\}$, is

$$B_x^{n+1}(i_b, j, k) = B_x^n(i_b, j, k) + \frac{\Delta t}{2} \left[\frac{E_y^{n+\frac{1}{2}}(i_b, j, k+1) - E_y^{n+\frac{1}{2}}(i_b, j, k)}{\Delta z} - \frac{E_z^{n+1}(i_b, j+1, k) - E_z^{n+1}(i_b, j, k)}{\Delta y} \right]. \quad (5.14)$$

Since the PEC condition gives

$$E_y(i_b, j, k) = 0, \quad E_z(i_b, j, k) = 0 \quad (5.15)$$

for all j and k , it follows that

$$B_x^{n+1}(i_b, j, k) = B_x^n(i_b, j, k). \quad (5.16)$$

Therefore, if

$$B_x^0(i_b, j, k) = 0, \quad (5.17)$$

then

$$B_x^n(i_b, j, k) = 0 \quad (5.18)$$

is already satisfied for all n .

6 Perfect Matching Layer Boundary Conditions

Here we derive the convolutional PML boundary conditions for the ADI scheme from [?].

6.1 First ADI Half-Step

E_x and H_z Using the same formulation as (3.9), the PML tridiagonal system is

$$\begin{aligned}
-\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} &= p_{x1} E_{x,i+\frac{1}{2},j,k}^n \\
&+ \frac{1}{\kappa_{y,i+\frac{1}{2},j,k}} \Delta y \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
&- \frac{1}{\kappa_{z,i+\frac{1}{2},j,k}} \Delta z \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\
&+ \frac{q_{x1}}{\kappa_{y,i+\frac{1}{2},j,k} \kappa_{x,i+\frac{1}{2},j-\frac{1}{2},k}} \Delta x \Delta y \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\
&- \frac{r_{x1}}{\kappa_{y,i+\frac{1}{2},j,k} \kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k}} \Delta x \Delta y \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\
&+ \frac{r_{x1}}{\kappa_{y,i+\frac{1}{2},j,k}} \Delta y \left(\psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^n - \psi_{hzy,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\
&- \frac{q_{x1}}{\kappa_{y,i+\frac{1}{2},j,k}} \Delta y \left(\psi_{hzy,i+\frac{1}{2},j-\frac{1}{2},k}^n - \psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^n \right) \\
&+ \left(\psi_{exy,i+\frac{1}{2},j,k}^n - \psi_{exz,i+\frac{1}{2},j,k}^n \right).
\end{aligned} \tag{6.1}$$

Here

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\kappa_{y,i+\frac{1}{2},j,k} \kappa_{y,i+\frac{1}{2},j-\frac{1}{2},k} \Delta y^2}, \tag{6.2}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{6.3}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\kappa_{y,i+\frac{1}{2},j,k} \kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k} \Delta y^2}, \tag{6.4}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{6.5}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{6.6}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{6.7}$$

and the PML magnetic update can be written as

$$\begin{aligned}
H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} &= H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n + \frac{r_{x1}}{\kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k}} \Delta y \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\
&- \frac{r_{x1}}{\kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k}} \Delta x \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\
&+ r_{x1} \psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^n - r_{x1} \psi_{hzy,i+\frac{1}{2},j-\frac{1}{2},k}^n.
\end{aligned} \tag{6.8}$$

E_y and H_x Using the same formulation as (3.24), the PML tridiagonal system is

$$\begin{aligned}
-\alpha_{y1} E_{y,i,j+\frac{1}{2},k-1}^{n+\frac{1}{2}} + \beta_{y1} E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - \gamma_{y1} E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} &= p_{y1} E_{y,i,j+\frac{1}{2},k}^n \\
&+ \frac{1}{\kappa_{z,i,j+\frac{1}{2},k} \Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^n \right) \\
&- \frac{1}{\kappa_{x,i,j+\frac{1}{2},k} \Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^n \right) \\
&+ \frac{q_{y1}}{\kappa_{z,i,j+\frac{1}{2},k} \kappa_{y,i,j+\frac{1}{2},k-\frac{1}{2}} \Delta y \Delta z} \left(E_{z,i,j+1,k-\frac{1}{2}}^n - E_{z,i,j,k-\frac{1}{2}}^n \right) \\
&- \frac{r_{y1}}{\kappa_{z,i,j+\frac{1}{2},k} \kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta y \Delta z} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right) \\
&+ \frac{r_{y1}}{\kappa_{z,i,j+\frac{1}{2},k} \Delta z} \left(\psi_{h x z,i,j+\frac{1}{2},k+\frac{1}{2}}^n - \psi_{h x y,i,j+\frac{1}{2},k+\frac{1}{2}}^n \right) \\
&- \frac{q_{y1}}{\kappa_{z,i,j+\frac{1}{2},k} \Delta z} \left(\psi_{h x z,i,j+\frac{1}{2},k-\frac{1}{2}}^n - \psi_{h x y,i,j+\frac{1}{2},k-\frac{1}{2}}^n \right) \\
&+ \left(\psi_{e y z,i,j+\frac{1}{2},k}^n - \psi_{e y x,i,j+\frac{1}{2},k}^n \right).
\end{aligned} \tag{6.9}$$

Here

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\kappa_{z,i,j+\frac{1}{2},k} \kappa_{z,i,j+\frac{1}{2},k-\frac{1}{2}} \Delta z^2}, \tag{6.10}$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \tag{6.11}$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\kappa_{z,i,j+\frac{1}{2},k} \kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta z^2}, \tag{6.12}$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{6.13}$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \tag{6.14}$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{6.15}$$

and the PML magnetic update can be written as

$$\begin{aligned}
H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} &= H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n + \frac{r_{y1}}{\kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
&- \frac{r_{y1}}{\kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^n - E_{z,i,j,k+\frac{1}{2}}^n \right) \\
&+ r_{y1} \psi_{h x z,i,j+\frac{1}{2},k+\frac{1}{2}}^n - r_{y1} \psi_{h x y,i,j+\frac{1}{2},k+\frac{1}{2}}^n.
\end{aligned} \tag{6.16}$$

E_z and H_y Using the same formulation as (3.39), the PML tridiagonal system is

$$\begin{aligned}
-\alpha_{z1} E_{z,i-1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \beta_{z1} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \gamma_{z1} E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} &= p_{z1} E_{z,i,j,k+\frac{1}{2}}^n \\
&+ \frac{1}{\kappa_{x,i,j,k+\frac{1}{2}} \Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^n \right) \\
&- \frac{1}{\kappa_{y,i,j,k+\frac{1}{2}} \Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^n - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^n \right) \\
&+ \frac{q_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}} \kappa_{z,i-\frac{1}{2},j,k+\frac{1}{2}} \Delta z \Delta x} \left(E_{x,i-\frac{1}{2},j,k+1}^n - E_{x,i-\frac{1}{2},j,k}^n \right) \\
&- \frac{r_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}} \kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}} \Delta z \Delta x} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right) \\
&+ \frac{r_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}} \Delta x} \left(\psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^n - \psi_{hyx,i-\frac{1}{2},j,k+\frac{1}{2}}^n \right) \\
&- \frac{q_{z1}}{\kappa_{x,i,j,k+\frac{1}{2}} \Delta x} \left(\psi_{hyx,i-\frac{1}{2},j,k+\frac{1}{2}}^n - \psi_{hyx,i-\frac{1}{2},j,k+\frac{1}{2}}^n \right) \\
&+ \left(\psi_{ezx,i,j,k+\frac{1}{2}}^n - \psi_{ezy,i,j,k+\frac{1}{2}}^n \right).
\end{aligned} \tag{6.17}$$

Here

$$\alpha_{z1} = \frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\kappa_{x,i,j,k+\frac{1}{2}} \kappa_{x,i-\frac{1}{2},j,k+\frac{1}{2}} \Delta x^2}, \tag{6.18}$$

$$\beta_{z1} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z1} + \gamma_{z1}, \tag{6.19}$$

$$\gamma_{z1} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\kappa_{x,i,j,k+\frac{1}{2}} \kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \Delta x^2}, \tag{6.20}$$

$$p_{z1} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{6.21}$$

$$q_{z1} = D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{6.22}$$

$$r_{z1} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{6.23}$$

and the PML magnetic update can be written as

$$\begin{aligned}
H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} &= H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n + \frac{r_{z1}}{\kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&- \frac{r_{z1}}{\kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}} \Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^n - E_{x,i+\frac{1}{2},j,k}^n \right) \\
&+ r_{z1} \psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^n - r_{z1} \psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^n.
\end{aligned} \tag{6.24}$$

If $\kappa_s = 1$ and all PML auxiliary variables vanish, equations (6.1), (6.9), and (6.17) reduce to (3.9), (3.24), and (3.39), respectively. Likewise, (6.8), (6.16), and (6.24) reduce to (3.10), (3.25), and (3.40).

6.2 First PML Auxiliary Variable Half-Step

The PML auxiliary variables are advanced over the first half-step from time level n to $n + \frac{1}{2}$. In this subsection, $\psi_{e\alpha s}$ denotes the electric auxiliary variable for the $\partial/\partial s$ derivative in the E_α update, and $\psi_{h\alpha s}$ denotes the magnetic auxiliary variable for the $\partial/\partial s$ derivative in the H_α update. Thus, for example, ψ_{exy} and ψ_{exz} correspond to the y - and z -derivative auxiliary variables in the E_x equation, respectively.

The electric auxiliary variables are updated from the completed magnetic fields at time level $n + \frac{1}{2}$,

$$\psi_{exy,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = b_{y,i+\frac{1}{2},j,k} \psi_{exy,i+\frac{1}{2},j,k}^n + a_{y,i+\frac{1}{2},j,k} \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta y}, \quad (6.25a)$$

$$\psi_{exz,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = b_{z,i+\frac{1}{2},j,k} \psi_{exz,i+\frac{1}{2},j,k}^n + a_{z,i+\frac{1}{2},j,k} \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z}, \quad (6.25b)$$

$$\psi_{eyx,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} = b_{x,i,j+\frac{1}{2},k} \psi_{eyx,i,j+\frac{1}{2},k}^n + a_{x,i,j+\frac{1}{2},k} \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x}, \quad (6.25c)$$

$$\psi_{eyz,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} = b_{z,i,j+\frac{1}{2},k} \psi_{eyz,i,j+\frac{1}{2},k}^n + a_{z,i,j+\frac{1}{2},k} \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta z}, \quad (6.25d)$$

$$\psi_{ezx,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = b_{x,i,j,k+\frac{1}{2}} \psi_{ezx,i,j,k+\frac{1}{2}}^n + a_{x,i,j,k+\frac{1}{2}} \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x}, \quad (6.25e)$$

$$\psi_{ezy,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} = b_{y,i,j,k+\frac{1}{2}} \psi_{ezy,i,j,k+\frac{1}{2}}^n + a_{y,i,j,k+\frac{1}{2}} \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y}. \quad (6.25f)$$

The PML recursion coefficients are,

$$b_x = \exp \left[- \left(\frac{\sigma_x}{\kappa_x} + \alpha_x \right) \frac{\Delta t}{2\varepsilon_0} \right], \quad (6.26a)$$

$$b_y = \exp \left[- \left(\frac{\sigma_y}{\kappa_y} + \alpha_y \right) \frac{\Delta t}{2\varepsilon_0} \right], \quad (6.26b)$$

$$b_z = \exp \left[- \left(\frac{\sigma_z}{\kappa_z} + \alpha_z \right) \frac{\Delta t}{2\varepsilon_0} \right], \quad (6.26c)$$

$$a_x = \frac{\sigma_x}{\kappa_x (\sigma_x + \kappa_x \alpha_x)} (b_x - 1), \quad (6.26d)$$

$$a_y = \frac{\sigma_y}{\kappa_y (\sigma_y + \kappa_y \alpha_y)} (b_y - 1), \quad (6.26e)$$

$$a_z = \frac{\sigma_z}{\kappa_z (\sigma_z + \kappa_z \alpha_z)} (b_z - 1). \quad (6.26f)$$

Here σ_s , α_s , and κ_s are evaluated at the same location as the corresponding auxiliary variable and in the same derivative direction s . For a PML layer normal to direction s , let ρ_s be the distance from the inner PML interface into the PML, $L_s = N_{\text{pml}} \Delta s$ be the PML thickness, and m be the polynomial grading order. Then, for $0 \leq \rho_s \leq L_s$,

$$\sigma_s(\rho_s) = \sigma_{s,\text{max}} \left(\frac{\rho_s}{L_s} \right)^m, \quad (6.27)$$

$$\kappa_s(\rho_s) = 1 + (\kappa_{s,\text{max}} - 1) \left(\frac{\rho_s}{L_s} \right)^m, \quad (6.28)$$

$$\alpha_s(\rho_s) = \alpha_{s,\text{max}} \left(1 - \frac{\rho_s}{L_s} \right), \quad (6.29)$$

while outside the PML,

$$\sigma_s = 0, \quad \alpha_s = 0, \quad \kappa_s = 1, \quad a_s = 0. \quad (6.30)$$

The maximum conductivity may be chosen from a target reflection coefficient R as

$$\sigma_{s,\text{max}} = - \frac{(m+1) \ln R}{2\eta_0 L_s}, \quad \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}. \quad (6.31)$$

The magnetic auxiliary variables are updated from the completed electric fields at time level $n + \frac{1}{2}$. Both field updates use

the auxiliary variables at time level n before these recursions are applied,

$$\psi_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = b_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \psi_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}}^n + a_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \frac{E_{z,i,j+1,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta y}, \quad (6.32a)$$

$$\psi_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} = b_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \psi_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}}^n + a_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \frac{E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta z}, \quad (6.32b)$$

$$\psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = b_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^n + a_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}}{\Delta x}, \quad (6.32c)$$

$$\psi_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} = b_{z,i+\frac{1}{2},j,k+\frac{1}{2}} \psi_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}}^n + a_{z,i+\frac{1}{2},j,k+\frac{1}{2}} \frac{E_{x,i+\frac{1}{2},j,k+1}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta z}, \quad (6.32d)$$

$$\psi_{hzx,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = b_{x,i+\frac{1}{2},j+\frac{1}{2},k} \psi_{hzx,i+\frac{1}{2},j+\frac{1}{2},k}^n + a_{x,i+\frac{1}{2},j+\frac{1}{2},k} \frac{E_{y,i+1,j+\frac{1}{2},k}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}}}{\Delta x}, \quad (6.32e)$$

$$\psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} = b_{y,i+\frac{1}{2},j+\frac{1}{2},k} \psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^n + a_{y,i+\frac{1}{2},j+\frac{1}{2},k} \frac{E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}}{\Delta y}. \quad (6.32f)$$

The magnetic auxiliary variables $\psi_{h\alpha s}$ are stored in the unscaled recursive-convolution form; therefore the magnetic-field coefficient D_b is applied when they enter the H updates, at the same staggered location as the corresponding magnetic field.

6.3 Second ADI Half-Step

E_x and H_y Using the same formulation as (3.55), the PML tridiagonal system is

$$\begin{aligned} -\alpha_{x2} E_{x,i+\frac{1}{2},j,k-1}^{n+1} + \beta_{x2} E_{x,i+\frac{1}{2},j,k}^{n+1} - \gamma_{x2} E_{x,i+\frac{1}{2},j,k+1}^{n+1} &= p_{x2} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\ &+ \frac{1}{\kappa_{y,i+\frac{1}{2},j,k} \Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &- \frac{1}{\kappa_{z,i+\frac{1}{2},j,k} \Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &+ \frac{q_{x2}}{\kappa_{z,i+\frac{1}{2},j,k} \kappa_{x,i+\frac{1}{2},j,k-\frac{1}{2}} \Delta x \Delta z} \left(E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{r_{x2}}{\kappa_{z,i+\frac{1}{2},j,k} \kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \Delta x \Delta z} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &- \frac{r_{x2}}{\kappa_{z,i+\frac{1}{2},j,k} \Delta z} \left(\psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \psi_{hyx,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &+ \frac{q_{x2}}{\kappa_{z,i+\frac{1}{2},j,k} \Delta z} \left(\psi_{hzy,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} - \psi_{hzy,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &+ \left(\psi_{exy,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \psi_{exz,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right). \end{aligned} \quad (6.33)$$

Here

$$\alpha_{x2} = \frac{D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\kappa_{z,i+\frac{1}{2},j,k}\kappa_{z,i+\frac{1}{2},j,k-\frac{1}{2}}\Delta z^2}, \quad (6.34)$$

$$\beta_{x2} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x2} + \gamma_{x2}, \quad (6.35)$$

$$\gamma_{x2} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\kappa_{z,i+\frac{1}{2},j,k}\kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta z^2}, \quad (6.36)$$

$$p_{x2} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \quad (6.37)$$

$$q_{x2} = D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}, \quad (6.38)$$

$$r_{x2} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \quad (6.39)$$

and the PML magnetic update can be written as

$$\begin{aligned} H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} &= H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{r_{x2}}{\kappa_{x,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta x} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{r_{x2}}{\kappa_{z,i+\frac{1}{2},j,k+\frac{1}{2}}\Delta z} \left(E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1} \right) \\ &\quad + r_{x2}\psi_{h_{yx},i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - r_{x2}\psi_{h_{yz},i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}}. \end{aligned} \quad (6.40)$$

E_y and H_z Using the same formulation as (3.70), the PML tridiagonal system is

$$\begin{aligned} -\alpha_{y2}E_{y,i-1,j+\frac{1}{2},k}^{n+1} + \beta_{y2}E_{y,i,j+\frac{1}{2},k}^{n+1} - \gamma_{y2}E_{y,i+1,j+\frac{1}{2},k}^{n+1} &= p_{y2}E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \\ &\quad + \frac{1}{\kappa_{z,i,j+\frac{1}{2},k}\Delta z} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{1}{\kappa_{x,i,j+\frac{1}{2},k}\Delta x} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &\quad + \frac{q_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\kappa_{y,i-\frac{1}{2},j+\frac{1}{2},k}\Delta y\Delta x} \left(E_{x,i-\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i-\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{r_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k}\Delta y\Delta x} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{r_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\Delta x} \left(\psi_{h_{zy},i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - \psi_{h_{zx},i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &\quad + \frac{q_{y2}}{\kappa_{x,i,j+\frac{1}{2},k}\Delta x} \left(\psi_{h_{zy},i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - \psi_{h_{zx},i-\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &\quad + \left(\psi_{e_{yz},i,j+\frac{1}{2},k}^{n+\frac{1}{2}} - \psi_{e_{yx},i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right). \end{aligned} \quad (6.41)$$

Here

$$\alpha_{y2} = \frac{D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\kappa_{x,i,j+\frac{1}{2},k} \kappa_{x,i-\frac{1}{2},j+\frac{1}{2},k} \Delta x^2}, \quad (6.42)$$

$$\beta_{y2} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y2} + \gamma_{y2}, \quad (6.43)$$

$$\gamma_{y2} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\kappa_{x,i,j+\frac{1}{2},k} \kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k} \Delta x^2}, \quad (6.44)$$

$$p_{y2} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \quad (6.45)$$

$$q_{y2} = D_{b,i-\frac{1}{2},j+\frac{1}{2},k}, \quad (6.46)$$

$$r_{y2} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \quad (6.47)$$

and the PML magnetic update can be written as

$$\begin{aligned} H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} &= H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + \frac{r_{y2}}{\kappa_{y,i+\frac{1}{2},j+\frac{1}{2},k} \Delta y} \left(E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{r_{y2}}{\kappa_{x,i+\frac{1}{2},j+\frac{1}{2},k} \Delta x} \left(E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1} \right) \\ &\quad + r_{y2} \psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - r_{y2} \psi_{hzx,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}}. \end{aligned} \quad (6.48)$$

E_z and H_x Using the same formulation as (3.85), the PML tridiagonal system is

$$\begin{aligned} -\alpha_{z2} E_{z,i,j-1,k+\frac{1}{2}}^{n+1} + \beta_{z2} E_{z,i,j,k+\frac{1}{2}}^{n+1} - \gamma_{z2} E_{z,i,j+1,k+\frac{1}{2}}^{n+1} &= p_{z2} E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \\ &\quad + \frac{1}{\kappa_{x,i,j,k+\frac{1}{2}} \Delta x} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{1}{\kappa_{y,i,j,k+\frac{1}{2}} \Delta y} \left(H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &\quad + \frac{q_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}} \kappa_{z,i,j-\frac{1}{2},k+\frac{1}{2}} \Delta z \Delta y} \left(E_{y,i,j-\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{r_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}} \kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta z \Delta y} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{r_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}} \Delta y} \left(\psi_{hxx,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - \psi_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &\quad + \frac{q_{z2}}{\kappa_{y,i,j,k+\frac{1}{2}} \Delta y} \left(\psi_{hxx,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - \psi_{hxy,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\ &\quad + \left(\psi_{exx,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - \psi_{exy,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right). \end{aligned} \quad (6.49)$$

Here

$$\alpha_{z2} = \frac{D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\kappa_{y,i,j,k+\frac{1}{2}} \kappa_{y,i,j-\frac{1}{2},k+\frac{1}{2}} \Delta y^2}, \quad (6.50)$$

$$\beta_{z2} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z2} + \gamma_{z2}, \quad (6.51)$$

$$\gamma_{z2} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\kappa_{y,i,j,k+\frac{1}{2}} \kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta y^2}, \quad (6.52)$$

$$p_{z2} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \quad (6.53)$$

$$q_{z2} = D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}, \quad (6.54)$$

$$r_{z2} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \quad (6.55)$$

and the PML magnetic update can be written as

$$\begin{aligned} H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} &= H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + \frac{r_{z2}}{\kappa_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta z} \left(E_{y,i,j+\frac{1}{2},k+1}^{n+\frac{1}{2}} - E_{y,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\ &\quad - \frac{r_{z2}}{\kappa_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \Delta y} \left(E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1} \right) \\ &\quad + r_{z2} \psi_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - r_{z2} \psi_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}. \end{aligned} \quad (6.56)$$

If $\kappa_s = 1$ and all PML auxiliary variables vanish, equations (6.33), (6.41), and (6.49) reduce to (3.55), (3.70), and (3.85), respectively. Likewise, (6.40), (6.48), and (6.56) reduce to (3.56), (3.71), and (3.86).

6.4 Second PML Auxiliary Variable Half-Step

The PML auxiliary variables are advanced over the second half-step from time level $n + \frac{1}{2}$ to $n + 1$. The notation for $\psi_{e\alpha s}$ and $\psi_{h\alpha s}$ is the same as in the first PML auxiliary-variable half-step.

The electric auxiliary variables are updated from the completed magnetic fields at time level $n + 1$,

$$\psi_{exy,i+\frac{1}{2},j,k}^{n+1} = b_{y,i+\frac{1}{2},j,k} \psi_{exy,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + a_{y,i+\frac{1}{2},j,k} \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+1}}{\Delta y}, \quad (6.57a)$$

$$\psi_{exz,i+\frac{1}{2},j,k}^{n+1} = b_{z,i+\frac{1}{2},j,k} \psi_{exz,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + a_{z,i+\frac{1}{2},j,k} \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+1}}{\Delta z}, \quad (6.57b)$$

$$\psi_{eyx,i,j+\frac{1}{2},k}^{n+1} = b_{x,i,j+\frac{1}{2},k} \psi_{eyx,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + a_{x,i,j+\frac{1}{2},k} \frac{H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} - H_{z,i-\frac{1}{2},j+\frac{1}{2},k}^{n+1}}{\Delta x}, \quad (6.57c)$$

$$\psi_{eyz,i,j+\frac{1}{2},k}^{n+1} = b_{z,i,j+\frac{1}{2},k} \psi_{eyz,i,j+\frac{1}{2},k}^{n+\frac{1}{2}} + a_{z,i,j+\frac{1}{2},k} \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} - H_{x,i,j+\frac{1}{2},k-\frac{1}{2}}^{n+1}}{\Delta z}, \quad (6.57d)$$

$$\psi_{ezx,i,j,k+\frac{1}{2}}^{n+1} = b_{x,i,j,k+\frac{1}{2}} \psi_{ezx,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + a_{x,i,j,k+\frac{1}{2}} \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}^{n+1}}{\Delta x}, \quad (6.57e)$$

$$\psi_{ezy,i,j,k+\frac{1}{2}}^{n+1} = b_{y,i,j,k+\frac{1}{2}} \psi_{ezy,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} + a_{y,i,j,k+\frac{1}{2}} \frac{H_{x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} - H_{x,i,j-\frac{1}{2},k+\frac{1}{2}}^{n+1}}{\Delta y}. \quad (6.57f)$$

The PML recursion coefficients are given by (6.26).

The magnetic auxiliary variables are updated from the completed electric fields at time level $n + 1$. Both field updates use

the auxiliary variables at time level $n + \frac{1}{2}$ before these recursions are applied,

$$\psi_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = b_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \psi_{hxy,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + a_{y,i,j+\frac{1}{2},k+\frac{1}{2}} \frac{E_{z,i,j+1,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta y}, \quad (6.58a)$$

$$\psi_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+1} = b_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \psi_{hxz,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + a_{z,i,j+\frac{1}{2},k+\frac{1}{2}} \frac{E_{y,i,j+\frac{1}{2},k+1}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta z}, \quad (6.58b)$$

$$\psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} = b_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \psi_{hyx,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + a_{x,i+\frac{1}{2},j,k+\frac{1}{2}} \frac{E_{z,i+1,j,k+\frac{1}{2}}^{n+1} - E_{z,i,j,k+\frac{1}{2}}^{n+1}}{\Delta x}, \quad (6.58c)$$

$$\psi_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+1} = b_{z,i+\frac{1}{2},j,k+\frac{1}{2}} \psi_{hyz,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} + a_{z,i+\frac{1}{2},j,k+\frac{1}{2}} \frac{E_{x,i+\frac{1}{2},j,k+1}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1}}{\Delta z}, \quad (6.58d)$$

$$\psi_{hzx,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} = b_{x,i+\frac{1}{2},j+\frac{1}{2},k} \psi_{hzx,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + a_{x,i+\frac{1}{2},j+\frac{1}{2},k} \frac{E_{y,i+1,j+\frac{1}{2},k}^{n+1} - E_{y,i,j+\frac{1}{2},k}^{n+1}}{\Delta x}, \quad (6.58e)$$

$$\psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^{n+1} = b_{y,i+\frac{1}{2},j+\frac{1}{2},k} \psi_{hzy,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} + a_{y,i+\frac{1}{2},j+\frac{1}{2},k} \frac{E_{x,i+\frac{1}{2},j+1,k}^{n+1} - E_{x,i+\frac{1}{2},j,k}^{n+1}}{\Delta y}. \quad (6.58f)$$

7 Dispersion Analysis

We analyze the homogeneous-vacuum ADI scheme for a wave propagating in the x -direction with

$$E_x = E_y = 0, \quad H_x = H_z = 0, \quad (7.1)$$

so that the only nontrivial fields are E_z and H_y . The other axis-aligned polarizations follow by a cyclic permutation of coordinates.

For simplicity, we consider the case of vacuum media. Set $\sigma = 0$, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$, and

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}. \quad (7.2)$$

The half-step coefficients (3.1) are then

$$C_a = 1, \quad C_b = \frac{\Delta t}{2\varepsilon_0}, \quad D_b = \frac{\Delta t}{2\mu_0}. \quad (7.3)$$

Define the coefficient γ by

$$\gamma := C_b D_b = \frac{(c\Delta t)^2}{4}. \quad (7.4)$$

Define the two staggered first-difference operators by

$$(\delta_x^E E_z)_{i+\frac{1}{2},j,k+\frac{1}{2}} := \frac{E_{z,i+1,j,k+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}}{\Delta x}, \quad (7.5a)$$

$$(\delta_x^H H_y)_{i,j,k+\frac{1}{2}} := \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x}. \quad (7.5b)$$

Thus δ_x^E maps the E_z grid to the H_y grid, while δ_x^H maps the H_y grid to the E_z grid. Their composition is the centered second difference

$$(\delta_x^2 E_z)_{i,j,k+\frac{1}{2}} := (\delta_x^H \delta_x^E E_z)_{i,j,k+\frac{1}{2}} = \frac{E_{z,i-1,j,k+\frac{1}{2}} - 2E_{z,i,j,k+\frac{1}{2}} + E_{z,i+1,j,k+\frac{1}{2}}}{\Delta x^2}. \quad (7.6)$$

Restricting (3.39)–(3.40) to this polarization and multiplying the electric equation by C_b , the first half-step is

$$(1 - \gamma \delta_x^2) E_z^{n+\frac{1}{2}} = E_z^n + C_b \delta_x^H H_y^n, \quad (7.7)$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b \delta_x^E E_z^{n+\frac{1}{2}}, \quad (7.8)$$

where E_z is implicit along x . In the second half-step, E_z is formally implicit along y . Because this mode has no y -dependence, that second-difference term vanishes and the update reduces to

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^{n+\frac{1}{2}}, \quad (7.9)$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b \delta_x^E E_z^{n+\frac{1}{2}}. \quad (7.10)$$

Combining (7.8) and (7.10), we have

$$H_y^{n+1} = H_y^n + 2D_b \delta_x^E E_z^{n+\frac{1}{2}} \quad (7.11)$$

At any time level m , write a spatial Fourier mode on the staggered Yee grid as

$$E_{z,i,j,k+\frac{1}{2}}^m = \hat{E}_z^m e^{ik_x i \Delta x}, \quad (7.12)$$

$$H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^m = \hat{H}_y^m e^{ik_x(i+\frac{1}{2})\Delta x}. \quad (7.13)$$

Plugging into (7.5) and (7.6) gives

$$(\delta_x^E E_z^m)_{i+\frac{1}{2},j,k+\frac{1}{2}} = iK_x \hat{E}_z^m e^{ik_x(i+\frac{1}{2})\Delta x}, \quad (7.14a)$$

$$(\delta_x^H H_y^m)_{i,j,k+\frac{1}{2}} = iK_x \hat{H}_y^m e^{ik_x i \Delta x}, \quad (7.14b)$$

$$(\delta_x^2 E_z^m)_{i,j,k+\frac{1}{2}} = -K_x^2 \hat{E}_z^m e^{ik_x i \Delta x}. \quad (7.14c)$$

where

$$K_x := \frac{2}{\Delta x} \sin\left(\frac{k_x \Delta x}{2}\right). \quad (7.15)$$

For brevity, define the derivative symbol and implicit denominator

$$s := iK_x, \quad D := 1 + \gamma K_x^2. \quad (7.16)$$

Substituting into (7.7) and using (7.14), the first-half electric update becomes

$$D \hat{E}_z^{n+\frac{1}{2}} = \hat{E}_z^n + C_b s \hat{H}_y^n. \quad (7.17)$$

Similarly for (7.8), we have

$$\hat{H}_y^{n+\frac{1}{2}} = \hat{H}_y^n + D_b s \hat{E}_z^{n+\frac{1}{2}}. \quad (7.18)$$

Substituting into (7.9) and using (7.14), the second-half electric update becomes

$$\hat{E}_z^{n+1} = \hat{E}_z^{n+\frac{1}{2}} + C_b s \hat{H}_y^{n+\frac{1}{2}}, \quad (7.19)$$

Similarly for (7.10), we have

$$\hat{H}_y^{n+1} = \hat{H}_y^n + 2D_b s \hat{E}_z^{n+\frac{1}{2}} \quad (7.20)$$

Solving (7.17) for the intermediate electric amplitude gives

$$\hat{E}_z^{n+\frac{1}{2}} = \frac{1}{D} \hat{E}_z^n + \frac{C_b s}{D} \hat{H}_y^n. \quad (7.21)$$

Substitution into (7.19) and (7.20), using $C_b D_b = \gamma$ and $s^2 = -K_x^2$, gives the one-step map

$$\begin{pmatrix} \hat{E}_z \\ \hat{H}_y \end{pmatrix}^{n+1} = G \begin{pmatrix} \hat{E}_z \\ \hat{H}_y \end{pmatrix}^n, \quad G = \frac{1}{D} \begin{pmatrix} 1 - \gamma K_x^2 & 2C_b s \\ 2D_b s & 1 - \gamma K_x^2 \end{pmatrix}. \quad (7.22)$$

A time-harmonic mode satisfies

$$\begin{pmatrix} \hat{E}_z \\ \hat{H}_y \end{pmatrix}^{n+1} = \lambda \begin{pmatrix} \hat{E}_z \\ \hat{H}_y \end{pmatrix}^n, \quad \lambda = e^{-i\omega \Delta t}. \quad (7.23)$$

The characteristic equation $\det(G - \lambda I) = 0$ yields

$$\lambda = \frac{1 - \gamma K_x^2 \pm 2i K_x \sqrt{\gamma}}{1 + \gamma K_x^2}. \quad (7.24)$$

Introduce the CFL number $S := c\Delta t/\Delta x$ and

$$\nu := S \sin\left(\frac{k_x \Delta x}{2}\right) = K_x \sqrt{\gamma}. \quad (7.25)$$

Then $\gamma K_x^2 = \nu^2$, and the eigenvalues simplify to

$$\lambda = \frac{1 \pm i\nu}{1 \mp i\nu}. \quad (7.26)$$

The forward-propagating branch ($\omega > 0$) is

$$\lambda = \frac{1 - i\nu}{1 + i\nu}, \quad (7.27)$$

for which $|\lambda| = 1$ indicating unconditional stability of the ADI scheme and there is no numerical dispersion. The phase angle is

$$\arg(\lambda) = -2 \arctan(\nu), \quad (7.28)$$

so the ADI numerical angular frequency is

$$\omega_{\text{ADI}}(k_x) = \frac{2}{\Delta t} \arctan\left(S \sin \frac{k_x \Delta x}{2}\right). \quad (7.29)$$

The corresponding numerical phase velocity is

$$v_{p,\text{ADI}}(k_x) = \frac{\omega_{\text{ADI}}(k_x)}{k_x}, \quad (7.30)$$

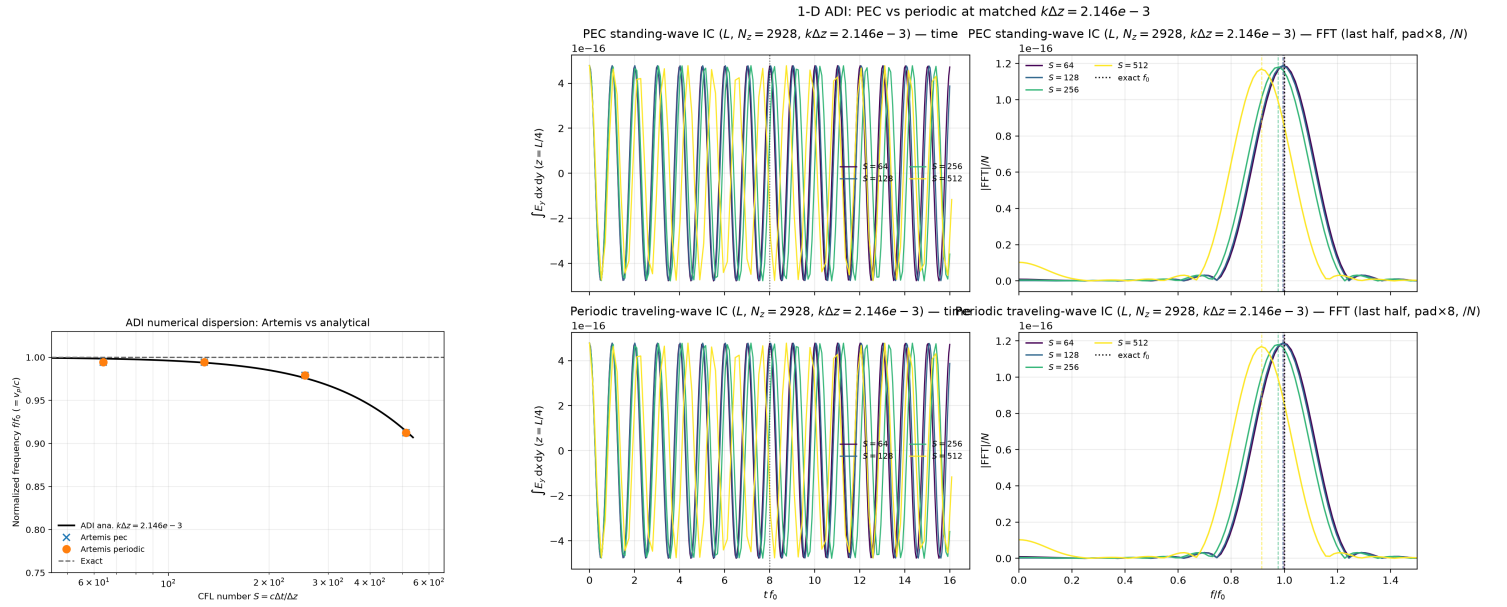
which generally differs from c and therefore exhibits numerical dispersion.

7.1 Numerical experiments

Initial-value problem verification The numerical dispersion relation was verified with a PEC standing wave and a periodic traveling wave. Both cases used the macroscopic vacuum ADI solver with the target resolution

$$k\Delta z \simeq 2.15 \times 10^{-3} \quad (7.31)$$

and CFL numbers $S = c\Delta t/\Delta z = 64, 128, 256$, and 512. In both cases $N_x = N_y = 8$. Each simulation was evolved for sixteen periods of the continuum frequency $f_0 = ck/(2\pi)$. The surface integral $\int E_y dx dy$ was sampled on a surface normal to z at $z = L_z/4$, approximately sixteen times per period. The dominant frequency was extracted from the last half of the signal using a Hann window, eightfold zero padding, and normalization by the number of samples.



(a) Normalized frequency versus CFL number.

(b) Time histories and corresponding frequency spectra.

Figure 7.1: Numerical verification of the ADI dispersion relation using PEC standing and periodic traveling waves.

Both tests used $L_z = 8 \mu\text{m}$ and $N_z = 2928$, corresponding to one wavelength across the domain:

$$k = \frac{2\pi}{L_z}, \quad k\Delta z = \frac{2\pi}{N_z} \simeq 2.146 \times 10^{-3}, \quad f_0 \simeq 3.75 \times 10^{13} \text{ Hz}. \quad (7.32)$$

For the PEC case, perfect-electric-conductor boundaries were imposed at $z = 0$ and $z = L_z$, with periodic boundaries in x and y , and the standing wave was initialized as

$$E_y = E_0 \sin(2\pi z/L_z), \quad \mathbf{B} = 0. \quad (7.33)$$

For the periodic case, periodic boundaries were imposed in all directions and the traveling wave was initialized as

$$E_y = E_0 \sin(2\pi z/L_z), \quad B_x = -E_y/c. \quad (7.34)$$

The measured FFT peak was divided by f_0 . For this fixed spatial mode the normalized frequency is also the normalized phase velocity, $f/f_0 = v_p/c$, and the analytic ADI prediction is

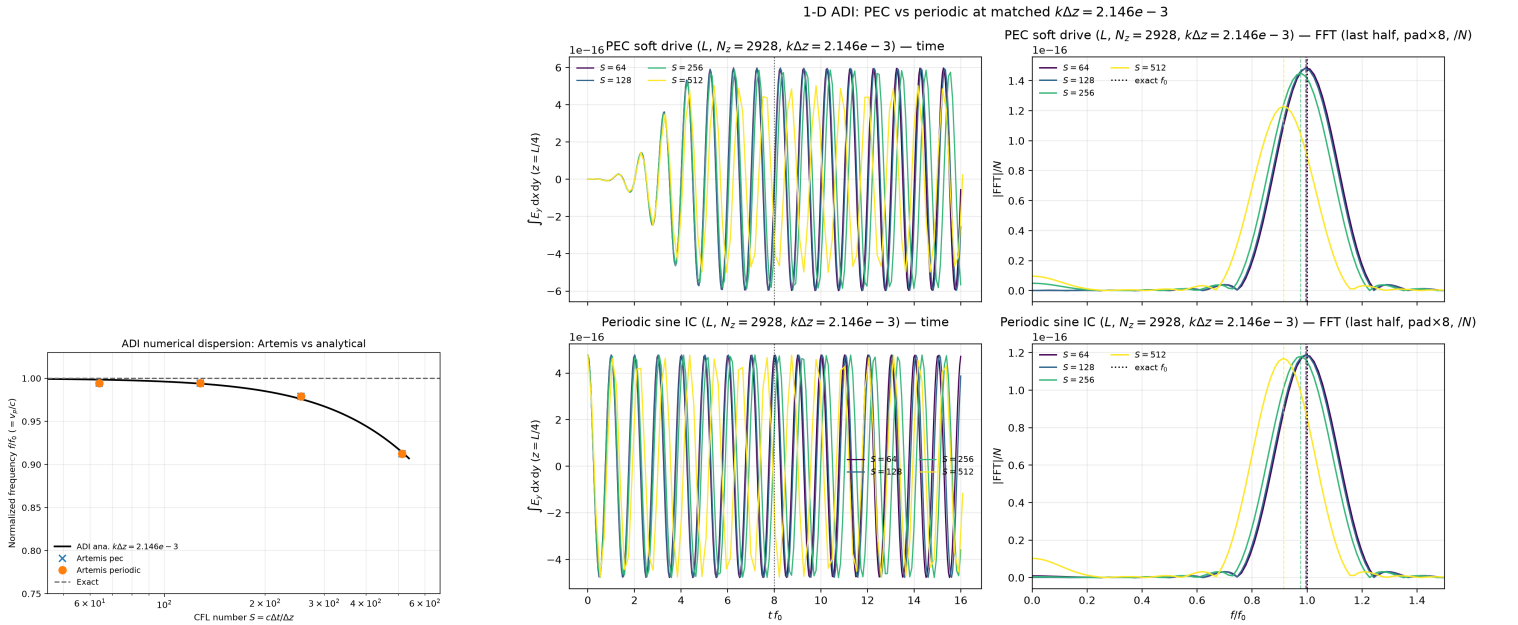
$$\frac{f_{\text{ADI}}}{f_0} = \frac{2 \arctan[S \sin(k\Delta z/2)]}{S k \Delta z}. \quad (7.35)$$

Figure 7.1 compares the measured frequencies with the analytic dispersion relation and shows the corresponding time histories and spectra. Both the periodic traveling-wave case and the PEC standing-wave case show good agreement with the analytical frequency shift, verifying the ADI dispersion relation. The agreement of both initial-value problems also indicates that the PEC boundary does not introduce observable additional error.

Soft excitation verification The dispersion relation was also tested by exciting the PEC cavity from initially zero fields. The geometry, material parameters, resolution, CFL numbers, run duration, and boundary conditions were identical to those of the PEC standing-wave test above. Only E_y was excited, using the soft source

$$S_{E_y}(z, t) = E_0 \frac{\Delta t}{T_P} \sin\left(\frac{2\pi z}{L_z}\right) \exp\left[-\frac{(t - 3T_P)^2}{2T_P^2}\right] \sin(2\pi f_0 t), \quad (7.36)$$

where $E_0 = 1$, $f_0 = c/L_z$, and $T_P = 1/f_0$. Thus the source has the spatial profile of the one-wavelength cavity mode and a Gaussian envelope centered at $3T_P$ with a width of one period. The factor $\Delta t/T_P$ keeps the total excitation approximately independent of the CFL number. No magnetic excitation was applied.



(a) Normalized frequency versus CFL number.

(b) Time histories and corresponding frequency spectra.

Figure 7.2: Numerical verification of the ADI dispersion relation using PEC soft excitation and periodic traveling waves.

The electric field was sampled at the mode antinode $z = L_z/4$ at approximately sixteen samples per period. As in the initial-value tests, the dominant frequency was extracted from the last half of the sixteen-period signal using a Hann-windowed, zero-padded Fourier transform. The measured frequencies were compared with (7.29) for $S = 64, 128, 256$, and 512 . A periodic traveling-wave initial-value problem on the same grid provided a reference.

The origin of the amplitude variation After the finite-duration source has vanished, the cavity oscillates at its intrinsic numerical frequency $\omega_{\text{ADI}}(S)$. The experiments show that $\omega_{\text{ADI}}(S)$ is independent of the excitation frequency ω_d . The residual amplitude, however, is determined by the spectral overlap of the source pulse with this cavity mode. For the Gaussian-modulated sinusoidal source used here, the positive-frequency contribution is approximately

$$A(S) \propto \exp\left[-\frac{T_P^2}{2}(\omega_{\text{ADI}}(S) - \omega_d)^2\right]. \quad (7.37)$$

Besides this detuning effect, the finite temporal sampling of the soft source changes the amount of drive injected into the cavity. The carrier is sampled once per time step with spacing $\Delta t = S\Delta z/c$. Representing each sampled value over one time-step interval gives the discrete-injection factor

$$\text{sinc}\left(\frac{\omega_d \Delta t}{2}\right) = \frac{\sin(\omega_d \Delta t/2)}{\omega_d \Delta t/2}. \quad (7.38)$$

This factor is separate from the ADI dispersion relation: it describes the loss of effective drive amplitude caused by resolving the carrier with a finite number of time steps. Combining the two effects gives the amplitude model

$$A(S) \propto \underbrace{\exp\left[-\frac{T_P^2}{2}(\omega_{\text{ADI}}(S) - \omega_d)^2\right]}_{\text{drive-mode detuning}} \underbrace{\text{sinc}\left(\frac{\omega_d \Delta t}{2}\right)}_{\text{discrete drive injection}}. \quad (7.39)$$

For the present grid, increasing the CFL number from 64 to 512 reduces the number of time steps per drive period from approximately 46 to 6. At $S = 512$, $\omega_d \Delta t/2 = \pi S/N_z \simeq 0.55$ and the injection factor is approximately 0.95. The detuning factor alone predicts an amplitude ratio of approximately 0.865, whereas inclusion of the discrete-injection factor gives 0.822, close to the measured value 0.827. This correction applies only to the externally driven case and is absent from the standing-wave initial-value problem.

Because $\omega_{\text{ADI}}(S)$ decreases as the CFL number increases, the response grows when this shift moves the numerical cavity frequency toward ω_d and decreases when it moves the cavity frequency away from ω_d . If the numerical frequency crosses the excitation frequency, the amplitude reaches a maximum near the crossing and the trend reverses. The observed CFL dependence therefore reflects drive-mode detuning, rather than numerical damping of the ADI mode; temporal sampling of the source may introduce an additional amplitude error at the largest CFL numbers.

Figure 7.3 compares the measured soft-excitation response with the detuning and discrete-injection models. The soft-excitation amplitude curve exactly matches our detuning and discrete-injection model.

PEC: soft drive vs standing IC ($k\Delta z = 2.146e - 3$)

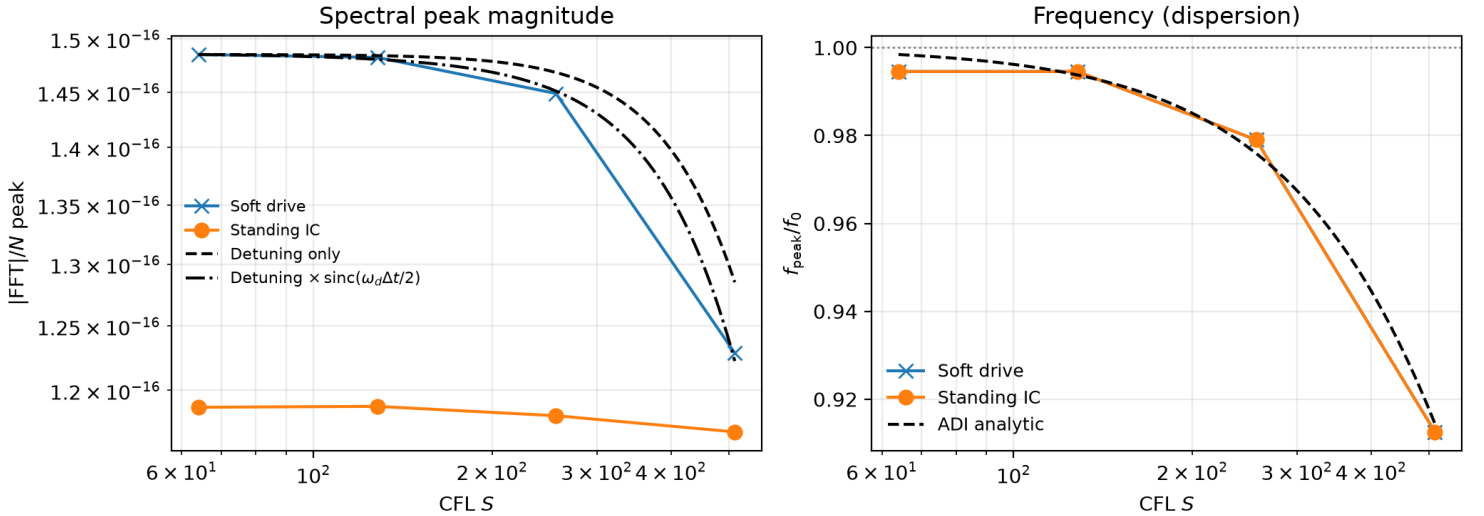


Figure 7.3: Amplitude and frequency of the PEC soft-excitation and standing-wave responses versus CFL number. The amplitude reference curves include the detuning-only model and the combined detuning and discrete-injection model.

8 External Excitation

External field excitations are prescribed by user-defined grid functions of space and time, together with a spatial flag that selects the excitation type at each Yee edge (or face). Following the Artemis convention,

- flag = 0: no excitation,
- flag = 1: hard source (Dirichlet), field equals the prescribed value,
- flag = 2: soft source, the prescribed value is *added* to the field update.

In the macroscopic ADI push these two families are treated differently: soft electric sources are built into both implicit electric systems, while the entire magnetic source increment is applied after the first magnetic half-step. Hard electric sources are not supported in the ADI electric solve.

8.1 Soft Electric Sources in the ADI Electric Update

Write $S_{E_x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}$ for the soft-source value assigned to the E_x edge at $(i+\frac{1}{2}, j, k)$ and the midpoint time level $n+\frac{1}{2}$ (and likewise for E_y and E_z at their staggered locations). Over a full step the intended soft contribution at that edge is split equally between the two ADI half-steps, with both contributions evaluated at the common midpoint $n+\frac{1}{2}$. The first-half electric update (3.5) is then modified to

$$E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} = C_{a, i+\frac{1}{2}, j, k} E_{x, i+\frac{1}{2}, j, k}^n + C_{b, i+\frac{1}{2}, j, k} [H_1] + \frac{1}{2} S_{E_x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}, \quad (8.1)$$

on edges where the E_x flag equals 2, and is unchanged elsewhere. The second half-step (3.51) is analogous,

$$E_{x, i+\frac{1}{2}, j, k}^{n+1} = C_{a, i+\frac{1}{2}, j, k} E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} + C_{b, i+\frac{1}{2}, j, k} [H_4] + \frac{1}{2} S_{E_x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}. \quad (8.2)$$

The same construction applies componentwise to E_y and E_z . The magnetic updates themselves are unchanged by the electric source; only the electric right-hand side is affected.

8.2 Effect on the ADI Tridiagonal System

Dividing (8.1) by $C_{b, i+\frac{1}{2}, j, k}$ and substituting the magnetic update as in Section 3 yields the same left-hand side as (3.9). On soft-source edges the only modification is an additional known term on the right-hand side:

$$\begin{aligned} -\alpha_{x1} E_{x, i+\frac{1}{2}, j-1, k}^{n+\frac{1}{2}} + \beta_{x1} E_{x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x, i+\frac{1}{2}, j+1, k}^{n+\frac{1}{2}} &= p_{x1} E_{x, i+\frac{1}{2}, j, k}^n \\ &+ \frac{1}{\Delta y} \left(H_{z, i+\frac{1}{2}, j+\frac{1}{2}, k}^n - H_{z, i+\frac{1}{2}, j-\frac{1}{2}, k}^n \right) \\ &- \frac{1}{\Delta z} \left(H_{y, i+\frac{1}{2}, j, k+\frac{1}{2}}^n - H_{y, i+\frac{1}{2}, j, k-\frac{1}{2}}^n \right) \\ &+ \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y, i+1, j-\frac{1}{2}, k}^n - E_{y, i, j-\frac{1}{2}, k}^n \right) \\ &- \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y, i+1, j+\frac{1}{2}, k}^n - E_{y, i, j+\frac{1}{2}, k}^n \right) \\ &+ \frac{1}{2C_{b, i+\frac{1}{2}, j, k}} S_{E_x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}}, \end{aligned} \quad (8.3)$$

with the same coefficients $\alpha_{x1}, \beta_{x1}, \gamma_{x1}, p_{x1}, q_{x1}, r_{x1}$ as in Section 3. Away from soft-source edges the final term is omitted. In the implementation this contribution is the increment

$$\text{rhs}_{i+\frac{1}{2}, j, k} += \frac{1}{2} S_{E_x, i+\frac{1}{2}, j, k}^{n+\frac{1}{2}} / C_{b, i+\frac{1}{2}, j, k} \quad (8.4)$$

added after the curl terms have been assembled. The E_y and E_z first-half systems are modified in exactly the same way.

The second half-step is similar. For example, the E_x system is

$$\begin{aligned}
-\alpha_{x2} E_{x,i+\frac{1}{2},j,k-1}^{n+1} + \beta_{x2} E_{x,i+\frac{1}{2},j,k}^{n+1} - \gamma_{x2} E_{x,i+\frac{1}{2},j,k+1}^{n+1} &= p_{x2} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} \\
&+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^{n+\frac{1}{2}} - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^{n+\frac{1}{2}} \right) \\
&- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^{n+\frac{1}{2}} - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&+ \frac{q_{x2}}{\Delta x \Delta z} \left(E_{z,i+1,j,k-\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k-\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&- \frac{r_{x2}}{\Delta x \Delta z} \left(E_{z,i+1,j,k+\frac{1}{2}}^{n+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}^{n+\frac{1}{2}} \right) \\
&+ \frac{1}{2C_{b,i+\frac{1}{2},j,k}} S_{E_x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}}.
\end{aligned} \tag{8.5}$$

Thus, after the second-half curl terms have been assembled, the normalized right-hand side is changed by

$$\text{rhs}_{i+\frac{1}{2},j,k} += \frac{1}{2} S_{E_x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} / C_{b,i+\frac{1}{2},j,k}. \tag{8.6}$$

The E_y and E_z second-half systems are modified in exactly the same way.

Consequently a hard electric source (flag = 1), which would require constraining the unknown E itself rather than adding to the right-hand side, is rejected in the ADI electric path.

8.3 Magnetic Sources After the First Half-Step

After the first ADI half-step has produced $\mathbf{E}^{n+\frac{1}{2}}$ and $\mathbf{B}^{n+\frac{1}{2}}$, an external magnetic excitation is applied directly to the magnetic field using the source sample evaluated at the midpoint $n + \frac{1}{2}$. The entire full-step source increment is applied at this point so that it enters the second electric solve. Thus, a soft magnetic source (flag = 2) contributes

$$B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} \leftarrow B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + S_{B_x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}, \tag{8.7}$$

while a hard magnetic source (flag = 1) overwrites

$$B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} \leftarrow S_{B_x,i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}}. \tag{8.8}$$

No additional magnetic excitation is applied after the second half-step. The same rules apply to B_y and B_z at their staggered faces. Unlike the electric soft source, this magnetic update is an explicit modification of the magnetic field rather than a term in a tridiagonal electric system; applying it after the first magnetic half-step allows it to affect the second electric solve.

8.4 Summary of the ADI Excitation Path

Over one full step Δt , the macroscopic ADI scheme therefore proceeds as follows.

1. Form and solve the first-half electric systems with soft electric sources $S_E^{n+\frac{1}{2}}$ added as $\frac{1}{2} S_E^{n+\frac{1}{2}}$ (equivalently $\frac{1}{2} S_E^{n+\frac{1}{2}} / C_b$ on the normalized RHS).
2. Update $\mathbf{B}$ to $n + \frac{1}{2}$ from the ADI magnetic formulae.
3. Apply the entire magnetic excitation to $\mathbf{B}^{n+\frac{1}{2}}$ (soft: add $S_B^{n+\frac{1}{2}}$; hard: set equal to $S_B^{n+\frac{1}{2}}$).
4. Form and solve the second-half electric systems with the same midpoint soft electric sources added as $\frac{1}{2} S_E^{n+\frac{1}{2}}$.
5. Update $\mathbf{B}$ to $n + 1$ without applying another magnetic excitation.

Hard electric sources are not part of this path. Soft electric sources never overwrite E ; they only augment the Ampère right-hand side of each half-step, consistently with the half-step coefficients C_a and C_b of (3.1).

9 Lumped Inductance

Then, we add a lumped inductor to the ADI electric update. A lumped inductor enters Ampère's law as an additional current density $\mathbf{J}_L$,

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} + \mathbf{J}_L \quad (9.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \quad (9.1b)$$

Here, we consider the x -directed inductor. The y - and z -directed inductors are analogous. For an x -directed edge of length Δx and face area $\Delta y \Delta z$, with total inductance L_x assigned to that edge,

$$V = E_x \Delta x, \quad I = J_{L,x} \Delta y \Delta z. \quad (9.2)$$

The inductor follows the relationship

$$V = L_x \frac{\partial I}{\partial t}. \quad (9.3)$$

Substituting the current and voltage relationships into the inductor constitutive relation, we get

$$E_x \Delta x = L_x \Delta y \Delta z \frac{\partial J_{L,x}}{\partial t}. \quad (9.4)$$

Define the inductor coupling constant

$$\kappa_x = \frac{\Delta x}{\Delta y \Delta z L_x}, \quad (9.5)$$

and the constitutive relation becomes

$$\frac{\partial J_{L,x}}{\partial t} = \kappa_x E_x. \quad (9.6)$$

Over a half step of size $\Delta t/2$, discretize (9.6) by the trapezoidal rule,

$$\frac{J_{L,x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - J_{L,x,i+\frac{1}{2},j,k}^n}{\Delta t/2} = \kappa_{x,i+\frac{1}{2},j,k} \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n}{2}. \quad (9.7)$$

Similarly as (3.2), discretize (9.1a) with field values averaged over the half step:

$$\varepsilon_{i+\frac{1}{2},j,k} \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^n}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k} \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n}{2} + \frac{J_{L,x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + J_{L,x,i+\frac{1}{2},j,k}^n}{2} = [H_1], \quad (9.8)$$

where $[H_1]$ is unchanged from (3.3). From (9.7),

$$\frac{J_{L,x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + J_{L,x,i+\frac{1}{2},j,k}^n}{2} = J_{L,x,i+\frac{1}{2},j,k}^n + \frac{\Delta t}{8} \kappa_{x,i+\frac{1}{2},j,k} (E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n). \quad (9.9)$$

Substituting into (9.8) and collecting unknowns gives

$$E_x^{n+\frac{1}{2}} = \frac{\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} - \frac{\Delta t}{8} \kappa_x}{\frac{\Delta t}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8} \kappa_x} E_x^n + \frac{1}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8} \kappa_x} ([H_1] - J_{L,x}^n). \quad (9.10)$$

Define the inductor-modified half-step coefficients

$$C_a^L = \frac{\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} - \frac{\Delta t}{8} \kappa_x}{\frac{\Delta t}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8} \kappa_x}, \quad C_b^L = \frac{1}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8} \kappa_x}. \quad (9.11)$$

When $\kappa_x = 0$ these reduce to (3.1). The electric update is then

$$E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = C_{a,i+\frac{1}{2},j,k}^L E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k}^L ([H_1] - J_{L,x,i+\frac{1}{2},j,k}^n). \quad (9.12)$$

Thus, compared with (3.5), κ_x modifies the local diagonal coefficients C_a^L and C_b^L , while the known current J_L^n enters the right-hand side. After $E_x^{n+\frac{1}{2}}$ is obtained, rearrange (9.7) to advance $J_{L,x}$ by

$$J_{L,x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = J_{L,x,i+\frac{1}{2},j,k}^n + \frac{\Delta t}{4} \kappa_{x,i+\frac{1}{2},j,k} (E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n). \quad (9.13)$$

Rearranging (9.12) yields the same stencil as (3.9), but with $C_{a,i+\frac{1}{2},j,k} \rightarrow C_{a,i+\frac{1}{2},j,k}^L$ and $C_{b,i+\frac{1}{2},j,k} \rightarrow C_{b,i+\frac{1}{2},j,k}^L$, and an additional current term:

$$\begin{aligned} -\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1}^L E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} &= p_{x1}^L E_{x,i+\frac{1}{2},j,k}^n \\ &+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\ &- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\ &+ \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\ &- \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\ &- J_{L,x,i+\frac{1}{2},j,k}^n, \end{aligned} \quad (9.14)$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \quad (9.15)$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \quad (9.16)$$

$$\beta_{x1}^L = \frac{1}{C_{b,i+\frac{1}{2},j,k}^L} + \alpha_{x1} + \gamma_{x1}, \quad (9.17)$$

$$p_{x1}^L = \frac{C_{a,i+\frac{1}{2},j,k}^L}{C_{b,i+\frac{1}{2},j,k}^L}, \quad (9.18)$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \quad (9.19)$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \quad (9.20)$$

The magnetic update for $H_z^{n+\frac{1}{2}}$ is unchanged, and the current update is (9.13). The other directions, and the second half-step, are analogous.

10 Lumped Resistance

Then, we add a lumped resistor to the ADI electric update. A lumped resistor on a Yee edge is an Ohmic current in parallel with the bulk conductivity. Unlike the inductor, it adds no auxiliary unknown and no further timestep restriction: the resistance is absorbed into the edge conductivity already present in the ADI half-step coefficients, and both half-steps use that conductivity. Zero resistance is the input convention for no resistor, not a short circuit. The resistor enters Ampère's law as an additional current density $\mathbf{J}_R$,

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} + \mathbf{J}_R \quad (10.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \quad (10.1b)$$

Here, we consider the x -directed resistor. The y - and z -directed resistors are analogous. For an x -directed edge of length Δx and face area $\Delta y \Delta z$, with total resistance R_x assigned to that edge,

$$V = E_x \Delta x, \quad I = J_{R,x} \Delta y \Delta z. \quad (10.2)$$

The resistor follows the relationship

$$V = R_x I. \quad (10.3)$$

Substituting the current and voltage relationships into the resistor constitutive relation, we get

$$E_x \Delta x = R_x \Delta y \Delta z J_{R,x}. \quad (10.4)$$

When $R_x > 0$, define the resistor coupling

$$\sigma_{R,x} = \frac{\Delta x}{\Delta y \Delta z R_x}, \quad (10.5)$$

and the constitutive relation becomes

$$J_{R,x} = \sigma_{R,x} E_x. \quad (10.6)$$

When $R_x = 0$ there is no resistor, so $\sigma_{R,x} = 0$. The conductivity used by ADI is therefore

$$\sigma_{\text{eff},x} = \sigma + \begin{cases} \Delta x / (\Delta y \Delta z R_x), & R_x > 0, \\ 0, & R_x = 0, \end{cases} \quad (10.7)$$

where σ is the bulk conductivity.

11 Lumped Capacitance

Then, we add a lumped capacitor to the ADI electric update. A lumped capacitor on a Yee edge is a capacitive current in parallel with the bulk displacement current. Unlike the inductor, it adds no auxiliary unknown: the capacitance is absorbed into the edge permittivity already present in the ADI half-step coefficients, and both half-steps use that permittivity. Zero capacitance means no capacitor. The capacitor enters Ampère's law as an additional current density $\mathbf{J}_C$,

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} + \mathbf{J}_C \quad (11.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \quad (11.1b)$$

Here, we consider the x -directed capacitor. The y - and z -directed capacitors are analogous. For an x -directed edge of length Δx and face area $\Delta y \Delta z$, with nonnegative, time-independent total capacitance C_x assigned to that edge,

$$V = E_x \Delta x, \quad I = J_{C,x} \Delta y \Delta z. \quad (11.2)$$

The capacitor follows the relationship

$$I = C_x \frac{\partial V}{\partial t}. \quad (11.3)$$

Substituting the current and voltage relationships into the capacitor constitutive relation, we get

$$J_{C,x} \Delta y \Delta z = C_x \Delta x \frac{\partial E_x}{\partial t}. \quad (11.4)$$

Define the capacitor coupling

$$\varepsilon_{C,x} = \frac{C_x \Delta x}{\Delta y \Delta z}, \quad (11.5)$$

and the constitutive relation becomes

$$J_{C,x} = \varepsilon_{C,x} \frac{\partial E_x}{\partial t}. \quad (11.6)$$

When $C_x = 0$ there is no capacitor, so $\varepsilon_{C,x} = 0$. The permittivity used by ADI is therefore

$$\varepsilon_{\text{eff},x} = \varepsilon + \varepsilon_{C,x} = \varepsilon + \frac{C_x \Delta x}{\Delta y \Delta z}. \quad (11.7)$$

where ε is the bulk permittivity interpolated to the electric edge.

12 Josephson Junction

Then, we add a Josephson junction to the ADI electric update. A Josephson junction enters Ampère's law as an additional current density $\mathbf{J}_J$,

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} + \mathbf{J}_J \quad (12.1a)$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \quad (12.1b)$$

Here, we consider the x -directed junction. The y - and z -directed junctions are analogous. For an x -directed edge of length Δx and face area $\Delta y \Delta z$, with critical current $I_{c,x}$ assigned to that edge,

$$V = E_x \Delta x, \quad I = J_{J,x} \Delta y \Delta z. \quad (12.2)$$

The junction follows the relationships

$$I = I_{c,x} \sin \phi_x, \quad \frac{\partial \phi_x}{\partial t} = \frac{2e}{\hbar} V. \quad (12.3)$$

Here $e > 0$ is the elementary charge and ϕ_x is the gauge-invariant phase difference, oriented consistently with the edge voltage. The critical current $I_{c,x}$ is measured in amperes per selected edge, not in amperes per unit area. The voltage $E_x \Delta x$ is the one-edge approximation to the voltage integral. Substituting the current and voltage relationships into the junction constitutive relations, we get

$$J_{J,x} \Delta y \Delta z = I_{c,x} \sin \phi_x, \quad \frac{\partial \phi_x}{\partial t} = \frac{2e}{\hbar} \Delta x E_x. \quad (12.4)$$

Define the junction coupling constant and critical current density

$$\eta_x = \frac{2e}{\hbar} \Delta x, \quad j_{c,x} = \frac{I_{c,x}}{\Delta y \Delta z}, \quad (12.5)$$

and the constitutive relations become

$$\frac{\partial \phi_x}{\partial t} = \eta_x E_x, \quad J_{J,x} = j_{c,x} \sin \phi_x. \quad (12.6)$$

For the resistively and capacitively shunted junction (RCSJ), place the resistor and capacitor on the same edge. Their contributions are absorbed into $\sigma_{\text{eff},x}$ and $\varepsilon_{\text{eff},x}$ from (10.7) and (11.7). In the following electric equations, σ and ε denote these effective edge values; without shunts they denote the interpolated bulk values. No separate linear inductor is required for the Josephson branch.

Over a half step of size $\Delta t/2$, discretize (12.6) by the trapezoidal rule,

$$\frac{\phi_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \phi_{x,i+\frac{1}{2},j,k}^n}{\Delta t/2} = \eta_x \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n}{2}. \quad (12.7)$$

Similarly as (3.2), discretize (12.1a) with field values averaged over the half step:

$$\varepsilon_{i+\frac{1}{2},j,k} \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - E_{x,i+\frac{1}{2},j,k}^n}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k} \frac{E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n}{2} + \frac{J_{J,x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + J_{J,x,i+\frac{1}{2},j,k}^n}{2} = [H_1], \quad (12.8)$$

where $[H_1]$ is unchanged from (3.3). Define the averaged current as

$$\bar{J}_{J,x}^n = \frac{J_{J,x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + J_{J,x,i+\frac{1}{2},j,k}^n}{2}. \quad (12.9)$$

From (12.7), suppressing the spatial indices,

$$\bar{J}_{J,x}^n = \frac{j_{c,x}}{2} \left\{ \sin \phi_x^n + \sin \left[\phi_x^n + \frac{\Delta t}{4} \eta_x \left(E_x^{n+\frac{1}{2}} + E_x^n \right) \right] \right\}. \quad (12.10)$$

Here the superscript n labels the average over the half-step from n to $n + \frac{1}{2}$; $\bar{J}_{J,x}^n$ still depends on the unknown $E_x^{n+\frac{1}{2}}$ and is not the instantaneous current $J_{J,x}^n$.

12.1 Newton Update

Substituting (12.10) into (12.8) and collecting unknowns gives

$$E_x^{n+\frac{1}{2}} = \frac{\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}} E_x^n + \frac{[H_1] - \bar{J}_{J,x}^n}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}}. \quad (12.11)$$

Define the half-step coefficients with the effective material properties

$$C_a^N = \frac{\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}}, \quad C_b^N = \frac{1}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}}. \quad (12.12)$$

When $I_{c,x} = 0$ the Josephson current vanishes, and these coefficients give the existing RC electric update; without shunts they reduce to (3.1). The electric update is then

$$E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = C_{a,i+\frac{1}{2},j,k}^N E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k}^N \left([H_1] - \bar{J}_{J,x,i+\frac{1}{2},j,k}^n \right). \quad (12.13)$$

Thus, compared with (3.5), the averaged current enters the electric equation as a local nonlinear function of the unknown field. Its appearance on the right-hand side does not make it a known source. After $E_x^{n+\frac{1}{2}}$ is obtained, rearrange (12.7) to advance ϕ_x by

$$\phi_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = \phi_{x,i+\frac{1}{2},j,k}^n + \frac{\Delta t}{4} \eta_x \left(E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n \right). \quad (12.14)$$

The instantaneous current is then $J_{J,x}^{n+\frac{1}{2}} = j_{c,x} \sin \phi_x^{n+\frac{1}{2}}$. The phase is the persistent junction state; the current can be reconstructed from it.

Rearranging (12.13) yields the same stencil as (3.9), but with $C_{a,i+\frac{1}{2},j,k} \rightarrow C_{a,i+\frac{1}{2},j,k}^N$ and $C_{b,i+\frac{1}{2},j,k} \rightarrow C_{b,i+\frac{1}{2},j,k}^N$, and an additional nonlinear current term:

$$\begin{aligned} -\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} &= p_{x1}^N E_{x,i+\frac{1}{2},j,k}^n \\ &+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\ &- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\ &+ \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\ &- \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\ &- \bar{J}_{J,x,i+\frac{1}{2},j,k}^n, \end{aligned} \quad (12.15)$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \quad (12.16)$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \quad (12.17)$$

$$\beta_{x1}^N = \frac{1}{C_{b,i+\frac{1}{2},j,k}^N} + \alpha_{x1} + \gamma_{x1}, \quad (12.18)$$

$$p_{x1}^N = \frac{C_{a,i+\frac{1}{2},j,k}^N}{C_{b,i+\frac{1}{2},j,k}^N}, \quad (12.19)$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \quad (12.20)$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \quad (12.21)$$

To solve this nonlinear system, let s index the solve direction y at fixed i, k , let u_s denote the unknown $E_x^{n+\frac{1}{2}}$, and let f_s denote the right-hand side without the averaged-current term. A predictor based on the old current is obtained from

$$-\alpha_s u_{s-1}^{(0)} + \beta_s^N u_s^{(0)} - \gamma_s u_{s+1}^{(0)} = f_s - j_{c,s} \sin \phi_s^n. \quad (12.22)$$

At Newton iteration m , compute the trial phase from the saved old phase and trial field,

$$\phi_s^{(m)} = \phi_s^n + \frac{\Delta t}{4} \eta_x \left(u_s^{(m)} + E_s^n \right), \quad (12.23)$$

and evaluate the nonlinear residual

$$F_s^{(m)} = -\alpha_s u_{s-1}^{(m)} + \beta_s^N u_s^{(m)} - \gamma_s u_{s+1}^{(m)} + \frac{j_{c,s}}{2} \left(\sin \phi_s^n + \sin \phi_s^{(m)} \right) - f_s. \quad (12.24)$$

The averaged current depends only on the field at the same edge, so its derivative modifies only the diagonal:

$$d_s^{(m)} = \left. \frac{\partial \bar{J}_{J,s}^n}{\partial u_s} \right|_{u_s^{(m)}} = \frac{\Delta t}{8} \eta_x j_{c,s} \cos \phi_s^{(m)}. \quad (12.25)$$

Define the Jacobian matrix as the derivative of the residual with respect to the electric-field unknowns. Its entries at iteration m are

$$\mathcal{J}_{sr}^{(m)} = \left. \frac{\partial F_s}{\partial u_r} \right|_{\mathbf{u}^{(m)}} = -\alpha_s \delta_{r,s-1} + \left(\beta_s^N + d_s^{(m)} \right) \delta_{r,s} - \gamma_s \delta_{r,s+1}, \quad (12.26)$$

where $\delta_{r,s}$ is the Kronecker delta. Write the diagonal entry explicitly as

$$b_s^{(m)} = \beta_s^N + \frac{\Delta t}{8} \eta_x j_{c,s} \cos \left[\phi_s^n + \frac{\Delta t}{4} \eta_x \left(u_s^{(m)} + E_s^n \right) \right]. \quad (12.27)$$

For a line with M unconstrained unknowns and prescribed endpoint values eliminated into the right-hand side, the Jacobian is

$$\mathcal{J}^{(m)} = \begin{pmatrix} b_1^{(m)} & -\gamma_1 & 0 & \cdots & 0 \\ -\alpha_2 & b_2^{(m)} & -\gamma_2 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -\alpha_{M-1} & b_{M-1}^{(m)} & -\gamma_{M-1} \\ 0 & \cdots & 0 & -\alpha_M & b_M^{(m)} \end{pmatrix}. \quad (12.28)$$

For a periodic line with $M > 2$ unique unknowns, identify $u_0 = u_M$ and $u_{M+1} = u_1$. The two additional corner entries make the matrix cyclic tridiagonal:

$$\mathcal{J}_{1M}^{(m)} = -\alpha_1, \quad \mathcal{J}_{M1}^{(m)} = -\gamma_M. \quad (12.29)$$

The nonlinear contribution changes only the diagonal; the off-diagonal and periodic corner entries remain those of the ADI curl operator. In matrix form, the Newton correction satisfies

$$\mathcal{J}^{(m)} \delta \mathbf{u} = -\mathbf{F}(\mathbf{u}^{(m)}). \quad (12.30)$$

The Newton correction therefore follows from another tridiagonal solve,

$$-\alpha_s \delta u_{s-1} + \left(\beta_s^N + d_s^{(m)} \right) \delta u_s - \gamma_s \delta u_{s+1} = -F_s^{(m)}. \quad (12.31)$$

Update the trial field with a damping factor $0 < \lambda \leq 1$,

$$u_s^{(m+1)} = u_s^{(m)} + \lambda \delta u_s, \quad (12.32)$$

and recompute the trial phase using (12.23). The old field, old phase, and explicit right-hand side remain fixed throughout the iteration. Start with $\lambda = 1$ and reduce it if the residual does not decrease. Phase increments are never accumulated across Newton iterations. The current and its derivative are evaluated locally, but the Newton correction is coupled along each ADI line through the tridiagonal matrix; no full three-dimensional Newton system is formed.

The magnetic update for $H_z^{n+\frac{1}{2}}$ is unchanged, and the phase update is (12.14). The other directions, and the second half-step, are analogous. In each half-step, solve the electric system to nonlinear convergence before committing the phase and updating the magnetic field. The implicit magnetic response is already represented by the electric stencil. A failed nonlinear solve must not be accepted silently.

For smooth solutions and sufficiently tight nonlinear tolerances, this centered coupling retains second-order temporal accuracy. In the isolated lossless small-phase limit it reduces to the centered linear-inductor update. This does not establish unconditional stability of the full nonlinear spatial ADI scheme: the Newton diagonal contribution can be negative, large steps can cause convergence difficulties, and endpoint averaging does not conserve the nonlinear Josephson energy exactly. Timestep refinement and nonlinear energy checks remain necessary.

12.2 Taylor Update

To obtain a linear electric solve, expand the new sine about the known old phase, rather than about zero. Define the phase increment

$$\delta\phi_x^n = \phi_x^{n+\frac{1}{2}} - \phi_x^n = \frac{\Delta t}{4}\eta_x \left(E_x^{n+\frac{1}{2}} + E_x^n \right). \quad (12.33)$$

Taylor expansion gives

$$\sin(\phi_x^n + \delta\phi_x^n) = \sin\phi_x^n + \cos\phi_x^n \delta\phi_x^n + O((\delta\phi_x^n)^2). \quad (12.34)$$

Define the phase-dependent junction coupling constant

$$\kappa_{J,x}^n = \eta_x j_{c,x} \cos\phi_x^n, \quad (12.35)$$

and the averaged current becomes

$$\bar{J}_{J,x}^n = J_{J,x}^n + \frac{\Delta t}{8}\kappa_{J,x}^n \left(E_x^{n+\frac{1}{2}} + E_x^n \right) + O(\Delta t^2), \quad J_{J,x}^n = j_{c,x} \sin\phi_x^n. \quad (12.36)$$

Substituting into (12.8), neglecting the $O(\Delta t^2)$ current remainder, and collecting unknowns gives

$$E_x^{n+\frac{1}{2}} = \frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} - \frac{\Delta t}{8}\kappa_{J,x}^n E_x^n + \frac{[H_1] - J_{J,x}^n}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8}\kappa_{J,x}^n}. \quad (12.37)$$

Define the junction-modified half-step coefficients

$$C_a^J = \frac{\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2} - \frac{\Delta t}{8}\kappa_{J,x}^n}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8}\kappa_{J,x}^n}, \quad C_b^J = \frac{1}{\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8}\kappa_{J,x}^n}. \quad (12.38)$$

When $I_{c,x} = 0$ the Josephson current vanishes, and these coefficients give the existing RC electric update; without shunts they reduce to (3.1). The electric update is then

$$E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = C_{a,i+\frac{1}{2},j,k}^J E_{x,i+\frac{1}{2},j,k}^n + C_{b,i+\frac{1}{2},j,k}^J \left([H_1] - J_{J,x,i+\frac{1}{2},j,k}^n \right). \quad (12.39)$$

Thus, compared with (3.5), $\kappa_{J,x}^n$ modifies the local diagonal coefficients C_a^J and C_b^J , while the known current $J_{J,x}^n$ enters the right-hand side. Both are evaluated from the old phase and held fixed during the half-step electric solve, just as the linear-inductor coupling and old current are fixed in the lumped-inductance update. After $E_x^{n+\frac{1}{2}}$ is obtained, rearrange (12.7) to advance ϕ_x by

$$\phi_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} = \phi_{x,i+\frac{1}{2},j,k}^n + \frac{\Delta t}{4}\eta_x \left(E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} + E_{x,i+\frac{1}{2},j,k}^n \right). \quad (12.40)$$

The instantaneous current is then $J_{J,x}^{n+\frac{1}{2}} = j_{c,x} \sin\phi_x^{n+\frac{1}{2}}$. The phase is the persistent junction state; the current can be reconstructed from it.

Rearranging (12.39) yields the same stencil as (3.9), but with $C_{a,i+\frac{1}{2},j,k} \rightarrow C_{a,i+\frac{1}{2},j,k}^J$ and $C_{b,i+\frac{1}{2},j,k} \rightarrow C_{b,i+\frac{1}{2},j,k}^J$, and an additional known current term:

$$\begin{aligned} -\alpha_{x1} E_{x,i+\frac{1}{2},j-1,k}^{n+\frac{1}{2}} + \beta_{x1} E_{x,i+\frac{1}{2},j,k}^{n+\frac{1}{2}} - \gamma_{x1} E_{x,i+\frac{1}{2},j+1,k}^{n+\frac{1}{2}} &= p_{x1}^J E_{x,i+\frac{1}{2},j,k}^n \\ &+ \frac{1}{\Delta y} \left(H_{z,i+\frac{1}{2},j+\frac{1}{2},k}^n - H_{z,i+\frac{1}{2},j-\frac{1}{2},k}^n \right) \\ &- \frac{1}{\Delta z} \left(H_{y,i+\frac{1}{2},j,k+\frac{1}{2}}^n - H_{y,i+\frac{1}{2},j,k-\frac{1}{2}}^n \right) \\ &+ \frac{q_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j-\frac{1}{2},k}^n - E_{y,i,j-\frac{1}{2},k}^n \right) \\ &- \frac{r_{x1}}{\Delta x \Delta y} \left(E_{y,i+1,j+\frac{1}{2},k}^n - E_{y,i,j+\frac{1}{2},k}^n \right) \\ &- J_{J,x,i+\frac{1}{2},j,k}^n, \end{aligned} \quad (12.41)$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \quad (12.42)$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \quad (12.43)$$

$$\beta_{x1}^J = \frac{1}{C_{b,i+\frac{1}{2},j,k}^J} + \alpha_{x1} + \gamma_{x1}, \quad (12.44)$$

$$p_{x1}^J = \frac{C_{a,i+\frac{1}{2},j,k}^J}{C_{b,i+\frac{1}{2},j,k}^J}, \quad (12.45)$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \quad (12.46)$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \quad (12.47)$$

The magnetic update for $H_z^{n+\frac{1}{2}}$ is unchanged, and the phase update is (12.40). The other directions, and the second half-step, are analogous. In the second half-step, recompute $J_{J,x}^{n+\frac{1}{2}} = j_{c,x} \sin \phi_x^{n+\frac{1}{2}}$ and $\kappa_{J,x}^{n+\frac{1}{2}} = \eta_x j_{c,x} \cos \phi_x^{n+\frac{1}{2}}$ from the updated phase. Each electric component requires only one tridiagonal solve per half-step; no Newton iteration or predictor–corrector loop is needed. The instantaneous current is always reconstructed using the full sine of the updated phase, not accumulated using a linearized current evolution.

For smooth solutions, $\delta \phi_x^n = O(\Delta t)$, so the omitted term in (12.36) is $O(\Delta t^2)$ in current. Its contribution to the half-step field update is $O(\Delta t^3)$ because $C_b^J = O(\Delta t)$. Consequently this linearization retains second-order temporal consistency; second-order convergence also requires a stable update and sufficiently accurate linear solves. Expanding about the old phase does not require ϕ_x^n itself to be small, but does require the phase change over a half-step to be resolved.

This is the linearly implicit counterpart of the centered lumped-inductor update. In the small-phase limit, $\sin \phi_x \simeq \phi_x$ and $\cos \phi_x \simeq 1$, it reduces to the linear-inductor coupling with

$$\kappa_x = \eta_x j_{c,x} = \frac{\Delta x}{\Delta y \Delta z L_{J,x}}, \quad L_{J,x} = \frac{\hbar}{2e I_{c,x}}. \quad (12.48)$$

For the isolated lossless small-phase junction, this limit is the centered linear-inductor update and has no plasma-frequency timestep stability limit. This does not establish unconditional stability of the full nonlinear spatial ADI scheme. The phase-dependent coupling $\kappa_{J,x}^n$ can be negative when $\cos \phi_x^n < 0$, and a large timestep can make the modified diagonal unsuitable for the unpivoted tridiagonal solver. A conservative local condition for retaining its positive structure is

$$\frac{1}{C_b^J} = \frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2} + \frac{\Delta t}{8} \kappa_{J,x}^n > 0. \quad (12.49)$$

This condition is not a complete nonlinear stability criterion. The linearized update does not conserve the nonlinear Josephson energy exactly, and large phase increments can be inaccurate even when the linear system is well conditioned. Timestep refinement and nonlinear energy checks remain necessary.